Theoretical and Applied Climatology (2025) 156:541
https://doi.org/10.1007/s00704-025-05739-x

RESEARCH

The effects of Socio-economic conditions and land use pattern on land surface temperature and urban heat Island at saint martin's Island in Bangladesh

Md. Rakib Hasan Rony[1] · Farhana Akter Jannat[1] · Avishek Adhikari[3] · Shahorare Ahomed[2] · Md. Rabiul Haque[4] · Md. Zakiur Rahman[2] · Md. Mostafizur Rahman[2]

Received: 17 April 2025 / Accepted: 15 August 2025 / Published online: 27 September 2025
© The Author(s), under exclusive licence to Springer-Verlag GmbH Austria, part of Springer Nature 2025

Abstract

This study explores complex relationships among socio-economic dynamics, land use transformations, and thermal environmental changes on Saint Martin's Island employing an integrated methodology, combining primary household surveys with Landsat-8 remote sensing analysis. A comprehensive questionnaire survey conducted among 225 households indicated Limited Literacy levels and economic constraints. Specifically, 88% of families have fewer than three income earners with an average monthly income of 15,000 BDT, which drives land conversion decisions. 46% of respondents are engaged in tourism-related enterprises due to limited alternative livelihoods. These socio-economic pressures directly impact environmental vulnerability through accelerated land use changes. Multitemporal analysis of Landsat-8 OLI/TIRS data (30 m spatial resolution, 2018–2022) using Normalized Difference Vegetation Index (NDVI), Normalized Difference Built-up Index (NDBI), Land Surface Temperature (LST), and Urban Heat Island (UHI) demonstrated notable environmental deterioration. The findings indicate that households with lower literacy and income levels tend to convert agricultural and residential land into commercial use, resulting in vegetation decline, expansion of built-up area (80.28% increase in NDBI), and concerning thermal deterioration by the UHI intensity, which increases from 2.6 °C to 3.8 °C in the hotspot areas. Correlation analysis further revealed that socioeconomically vulnerable areas exhibit the strongest negative correlations between vegetation cover and thermal parameters (NDVI-LST: r = -0.576), whereas urban expansion shows a positive correlation with temperature increases (NDBI-LST: r=0.393). The northwestern region, characterized by the highest concentration of development driven by economic factors, experienced the most pronounced UHI intensification (1.9 °C), thereby illustrating how socio-economic constraints increase heat vulnerability and environmental degradation.

✉ Md. Mostafizur Rahman
mostafizur.ges@brur.ac.bd

Md. Rakib Hasan Rony
rakib.h.ronie@gmail.com

Farhana Akter Jannat
farhana01jannat@gmail.com

Avishek Adhikari
avishekadhikari99@gmail.com

Shahorare Ahomed
shahorareahomed@gmail.com

Md. Rabiul Haque
rabiuldps@du.ac.bd

Md. Zakiur Rahman
mzrahman@brur.ac.bd

1 Department of Environmental Science and Engineering, University of Science and Technology Beijing, Beijing, China

2 Department of Geography and Environmental Science, Begum Rokeya University, Rangpur 5400, Bangladesh

3 BUET-Japan Institute of Disaster Prevention and Urban Safety, University of Engineering and Technology, Dhaka, Bangladesh

4 Department of Population Sciences, University of Dhaka, Dhaka, Bangladesh

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

1 Introduction

A land use survey constitutes an indispensable procedure aimed at identifying the most suitable utilization of land resources (Yanbo et al. 2023). Such a survey assesses attributes such as soil properties, climatic conditions, hydrological features, geological characteristics, vegetation, and fauna within a specified area (Ara et al. 2021). The term 'land use' refers to human activities conducted on land, whereas 'land cover' pertains to the physical features present on the land surface (AbdelRahman 2023). By comparing historical and current cartographic data and gathering relevant information, a land use survey generates a comprehensive report. This report delineates the variability of land features and depicts the spatial distribution of different land units through mapping (Ara et al. 2021).

Saint Martin's Island is the sole coral island in Bangladesh, distinguished by its unique marine biotic resources (Das et al. 2022); this makes it the most biodiverse hotspot in the country and an exceptionally prestigious destination for travelers. Various marine sedimentary rocks, originating from the late Miocene to the present, are present on the island, which is situated in the Bay of Bengal (Nafi and Ahmed 2017). These rocks exhibit many features, including rocky platforms, sandy beaches, lagoons, dunes, marshes, and coral clusters. This ecosystem is enhanced by abundant, globally significant marine and terrestrial resources, such as the coexistence of corals, seaweed, algae, grasses, and mangroves (Ayyam et al. 2019). The island's geological composition consists of Holocene-era sand and shell fragments overlaying impermeable Tertiary shale and calcareous sandstone. This geological configuration results in predominantly saline groundwater, which exhibits seasonal fluctuations (Chowdhury et al. 1997). Despite being only 3 m above sea level, the tropical climate, characterized by substantial rainfall throughout the year, presents significant challenges (Saha et al. 2024). Classified as Am by the Köppen-Geiger system, the island experiences minimal relief during its short dry season (Saha et al. 2024).

The profile of Saint Martin's Island, as the sole coral island in Bangladesh, has been significantly elevated through increased global connectivity and tourism promotion. Despite the government's seasonal restrictions, annually, Saint Martin Island receives approximately 944,730 visits, representing 3.28% of Bangladesh's overnight trips, with daily tourist numbers rising to 3500–4000 during peak season, highlighting both its popularity and the environmental strain from unregulated tourism (Biswas et al. 2025). That is how the tourism sector of this area experienced a spike. The locals are investing their land and resources in building infrastructure to support tourism, which is one of the main reasons for changes in the island's land use pattern (Tania 2024). Saint Martin's Island, once limited to foreigners and non-resident Bangladeshis, now welcomes residents, increasing popularity as a tourist destination with daily trips from Chittagong and Cox's Bazar. However, conservation efforts are crucial to protect the island's coral reefs, endangered turtle species, and unique Narikel-Jinjira ecosystem (B. Roy 2020). Overfishing poses a significant threat, pushing fishermen to venture farther into the deep sea.

Environmental degradation is a major concern on Saint Martin's Island at present, largely due to the extensive construction of resorts and infrastructure to support tourists from across Bangladesh. This development leads to deforestation and environmental pollution (Ahammed et al. 2015). Reducing the number of trees and green spaces may lead to severe heat in this region. As an island, this could result in permanent ecological damage and the destruction of the island's environment, potentially causing the deterioration of the only coral island in Bangladesh (Islam et al. 2024a, 2024b). Additionally, changing land uses significantly impacts the vegetation index, build-up index, and surface temperatures (Ahmed et al. 2023).

Multiple studies have been conducted on Saint Martin's Island, highlighting the overexploitation of fish and shrimp resources, while another study warns of a shrinking shoreline posing a major threat to the island's future (Massey 2017). Coral reefs' degradation compounds these pollution concerns, identified by (Hossen and Sultana 2023) as a key factor in climate change. Their research also reported a significant shift in land use, with a large number of hotel and resort constructions leading to a 38% reduction in coral cover (Hossen and Sultana 2023). Satellite data analysis further reveals the negative impact of human settlements on vegetation health (B. Roy 2020), a trend of anthropogenic pressure consistent with findings from other sensitive ecosystems in Bangladesh, such as the Madhupur Sal Forest, where settlement encroachment has driven significant deforestation (Islam et al. 2023). These localized pressures on the island are compounded by regional climate change. This trend is particularly found in Bangladesh's major urban centers, where rapid, migration-driven growth has led to a significant increase in built-up areas at the expense of green space (Rashid et al. 2023), which in turn intensifies the UHI effect and creates public health challenges related to heat waves (M. Y. Islam et al. 2024a, 2024b); this makes the analysis of land use-driven thermal challenges on Saint Martin not just a localized issue, but part of a critical national environmental challenge.

Climate projections and recent observations indicate that regional warming is affecting Saint Martin's Island. Indeed, Bangladesh's climate has experienced significant warming over recent decades; one study observed an increase in mean annual temperatures of approximately 0.13 °C per

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

decade from 1949 to 2013 (Alam et al. 2023). Additionally, the waters of the Bay of Bengal surrounding the island frequently attain temperatures in the low 30 s °C, thereby intensifying storm activity. Coral islands such as Saint Martin's are recognized as particularly susceptible to climate-induced warming and rising sea levels (Alam et al. 2023). In essence, global climate change is plausibly contributing to the elevation of land surface temperatures on the island, alongside local land-use influences. Most of these studies predominantly focused on the land use pattern, along with land surface temperature (LST) and the urban heat island (UHI). However, none of the past studies examined the linkage between socio-economic conditions and changes in land use patterns with changes in LST and UHI values in the study island.

In the context of Bangladesh, while numerous studies have documented the impacts of urbanization on LST in major cities (Abdullah et al. 2022; Ahmed et al. 2013; Imran et al. 2021; Rashid et al. 2022), a persistent research gap exists in understanding these dynamics in ecologically sensitive coastal zones, specifically in Saint Martin's Island. Critically, the environmental consequences of the rapidly growing tourism sector, particularly its impact on local microclimates, remain significantly under-researched (Ara et al. 2021; Biswas et al. 2025; Zahra 2013). This gap is particularly acute for Saint Martin's Island, where the compounding pressures of global climate change and local, tourism-driven development are not fully understood. Firstly, it provides the first context-specific analysis for Saint Martin's Island that analytically links specific socio-economic drivers from household surveys to observed changes in LST and UHI. Secondly, its methodological novelty lies in its integrated mixed-methods approach, which combines quantitative remote sensing data with qualitative survey data to explain the causal mechanisms of environmental change. This integrated framework answers calls for more holistic socio-ecological systems (SES) research that can address complex environmental problems. Finally, the study offers policy-relevant novelty by providing scientific evidence that connects unregulated tourism development to specific thermal impacts, offering a crucial resource for sustainable land management in Bangladesh's vulnerable coastal regions. This contribution is key to understanding and mitigating the risks of irreversible ecological degradation on the island.

2 Study area and methods

2.1 Selection of the study area

Saint Martin's Island is situated in the Teknaf Upazila of Cox's Bazar district, 192 km south of the Port City of Chittagong and 109 km south of Cox's Bazar town. According to Google Earth Satellite Imagery, the northernmost point has a coordinate value of 20°38'12.66''N and 92°19'38.74''E. The southern tip's latitude and longitude values are 20°34'33.43''N and 92°20''13.19''E, as collected from Google Earth Satellite Image. The island has three major islands, Central Island being the biggest and commonly known as St. Martin's Island. St. Martin's Island possesses a diverse array of tree species, encompassing native mangroves such as Lumnitzera racemosa and introduced species, including coconut, pandanus, and casuarina, which are notably prevalent in coastal and dune regions (Upal 2015). A total of 78 plant species from 40 families have been documented, thereby emphasizing the island's ecological richness and highlighting the pressing need for conservation efforts in response to escalating environmental concerns pressures (Upal 2015). Figure 1 shows the map of Saint Martin's Island.

Approximately 3,700 individuals reside on th including four amphibian species, 28 reptile species, 130 bird species, and 20 mammal species (Ahammed et al. 2015). The island sustains four e island, predominantly comprising fishermen from 535 households. A total of 102 animal species are documented as inhabiting the island,months of tourism activity annually, spanning from November to February. Freshwater sources are limited to a few ponds and tube wells, which supply water for drinking and agricultural purposes. Despite being situated within a monsoon region, the local climate is notably influenced by the sea (Monir et al. 2023). The predominant vegetation consists of bamboo, betel nuts, and coconuts, with the Jinjira region exhibiting a higher density and abundance of coconut trees. The main agricultural produce includes watermelon, onions, and various types of rice (Ahammed et al. 2015). Recent remote sensing studies confirm the island's land cover is rapidly changing, making it highly vulnerable to thermal hazards. One study documented a dramatic 619% increase in built-up areas between 2005 and 2021, a period during which agricultural land, natural vegetation, and bare sand all decreased significantly (Ahmed et al. 2023). Another analysis, covering the period from 2005 to 2019, quantified this shift, finding that built-up areas grew by 175%, while agricultural land and coral reefs declined by 15% and 38%, respectively (Ara et al. 2021). This trend of development has continued in the most recent period, with a further 2.52% increase in developed areas observed between 2018 and 2024 (Biswas et al. 2025). The conversion of cooling landscapes to heat-absorbing surfaces justifies selecting Saint Martin's Island for studying the impacts on LST and UHI.

These socio-ecological characteristics of a study area are directly relevant to the study of LST and UHI (Ullah et al. 2025). The concentrated four-month tourism season of this

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

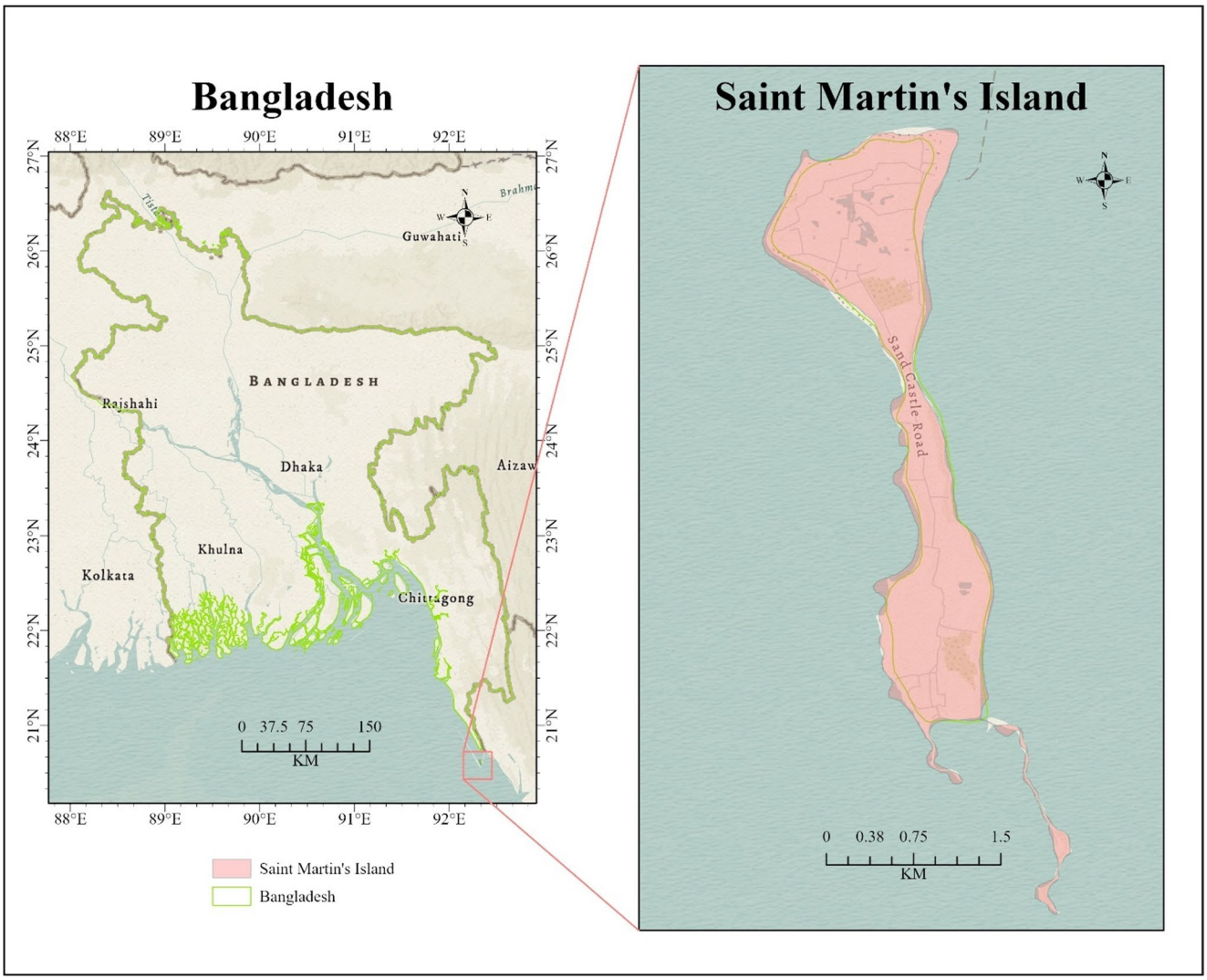


Fig. 1 Location map of the study area (Saint Martin's Island, Bangladesh)

island provides the intense economic impetus for converting land from its natural state to commercial use. The loss of traditional agricultural land and native vegetation, such as coconut groves, involves replacing surfaces that have natural cooling properties (via evapotranspiration and shading) with impervious, heat-absorbing materials like concrete and asphalt. Furthermore, the limited freshwater ponds act as localized cooling elements; their removal for construction would further intensify surface temperatures (Syafii et al. 2021). Therefore, Saint Martin's Island represents an ideal case study, a microcosm where these specific socio-ecological pressures are actively and rapidly transforming the island's thermal landscape (Biswas et al. 2025).

2.2 Data collection and methods

This study utilizes a combination of primary and secondary data sources to provide a comprehensive understanding of land use changes from 2018 to 2022, as well as potential future modifications within the study area. Primary data was collected via a questionnaire designed to capture respondents' socio-economic status and land use patterns. This comprehensive survey addresses land use, socio-economic, and environmental factors. It gathers demographic data, household characteristics, migration trends, and housing facilities. The survey also investigates land ownership, changes in land use over time, land values, and associated

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

challenges from 2018 to 2022. Furthermore, it encompasses information on income and expenditures, health-related aspects such as water sources and medical care, as well as environmental concerns including pollution, natural disasters, and social cohesion within the community. Data were collected through a consensual interview conducted with pencil and paper. Subsequently, the collected data were entered into an Excel spreadsheet, and the analyses were performed accordingly.

A purposive sampling technique was employed to select 225 households for the questionnaire survey. This non-probability sampling method was strategically chosen for its ability to identify and select 'information-rich' cases from which one can learn a great deal about issues of central importance to the purpose of the research (Etikan 2016; Patton 2015). The primary respondents were heads of households or long-term residents, targeted for their specific knowledge of their family's economic history and land use decisions over the study period. To ensure data representatives across the island, surveyors gathered responses from various locations, targeting areas with both significant recent development and long-standing residential zones. Data collection was performed with the informed consent of each participant.

Satellite imagery from Landsat-8 Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS) was obtained through the United States Geological Survey (USGS) Earth Explorer portal (https://earthexplorer.usgs.gov/) to derive spectral indices, including NDVI, NDBI, LST, and UHI intensity mapping (Roy et al. 2014; Wulder et al. 2022). The satellite data corresponded to path 135 and row 46 of the Worldwide Reference System-2 (WRS-2), with data acquisition occurring in December of both study years to ensure seasonal consistency and minimize phenological variability (Kafy et al. 2021; Loveland and Dwyer 2012). The spatial resolution of the acquired imagery was 30 m × 30 m for all spectral bands, except thermal infrared bands 10 and 11, which possess a native spatial resolution of 100 m × 100 m. Stringent quality control measures were implemented to ensure optimal atmospheric conditions, with all selected imagery maintaining cloud coverage below 5% to minimize atmospheric interference and ensure data reliability for subsequent analyses (Kafy et al. 2020; Foga et al. 2017). Tables 1 and 2 show the details of the data used in this study.

This study utilized Landsat 8 Collection 2 Level-2 science products, which the USGS provides as analysis-ready data (USGS 2021). The Level-2 data are processed to Surface Reflectance, having already been corrected for atmospheric effects such as aerosol scattering and absorption using the Land Surface Reflectance Code (LaSRC) (Vermote et al. 2016). Therefore, no manual atmospheric correction was necessary. To ensure the highest quality data was used for analysis, the included pixel quality assessment (QA) band was used to create a mask to remove pixels contaminated by clouds and cloud shadows, following the standard recommended procedures for this data product (Foga et al. 2017). The choice of December is also a practical response to regional climatic conditions; acquiring high-quality, cloud-free optical imagery in Bangladesh is exceptionally challenging during the monsoon season, making the dry season the most viable period for remote sensing analysis (Chander et al. 2009). Furthermore, this period aligns with the peak tourist season on Saint Martin's Island (November-February), allowing our analysis to capture the thermal environment when the impacts of tourism-related land use and infrastructure are at their maximum (Ara et al. 2021). The four-year interval was chosen to capture a period of significant and accelerated development on the island, allowing for a precise analysis of recent environmental changes.

Ground-based meteorological stations are unavailable on Saint Martin's Island due to its remote location and limited size. For validation purposes, this study utilized temperature data from the nearest Bangladesh Meteorological Department (BMD) station, located in Teknaf (20.861642, 92.294063), which is within the same upazila as Saint

Table 1 Landsat-8 satellite data acquisition details

Year	Landsat	Date of Acquisition	Sensor	Cloud Coverage	Spectral Resolution	Path and Row
2018	Landsat-8	24 December 2018	OLI_TIRS	< 5%	30 m × 30 m (spectral bands)	135/46
2022		11 December, 2022	Level 2		100 m × 100 m (thermal bands)	

Table 2 Landsat-8 bands used for analysis

Index	Bands Used	Band Name	Wavelength	Purpose
NDVI	Band 4 (Red)	Red	0.64–0.67 µm	Measures vegetation health
	Band 5 (NIR)	Near-Infrared	0.85–0.88 µm	
NDBI	Band 5 (NIR)	Near-Infrared	0.85–0.88 µm	Measures built-up areas
	Band 6 (SWIR 1)	Short-Wave Infrared 1	1.57–1.65 µm	
LST/UHI	Band 10 (TIRS)	Thermal Infrared	10.60–11.19 µm	Measures LST and UHI intensity

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

Martin's Island and situated approximately 32 km away. Data for December in both years were collected to facilitate validation with LST and UHI. The integration of primary field observations alongside secondary remote sensing data sources enabled a comprehensive multi-temporal analysis of LULC changes within the study area (Phan et al. 2020). This approach strengthens the analytical framework by integrating multiple data streams, providing a unique perspective on the complex spatiotemporal dynamics influencing land use changes and their impacts on regional socio-economic and environmental systems (Kafy et al. 2021). Geospatial data processing and index calculations were executed utilizing ArcMap 10.8.2 to derive NDVI, NDBI, LST, and UHI intensity metrics (Ahmed et al. 2013; Deng and Wu 2012). Advanced statistical analyses and algorithms for temporal change detection were employed within the Jupyter Notebook environment, utilizing the Python 3.9 programming language, along with various libraries for the quantitative evaluation of different analyses. Figure 2 shows the detailed flow of this research.

2.3 Socio-economic data analysis

This study employed both descriptive and inferential statistical methods to analyze the socio-economic characteristics of households on Saint Martin's Island. Data were collected through a survey and were analyzed descriptively and inferentially. Descriptive statistics summarized demographic trends, while inferential statistics assessed relationships between key variables. Chi-square tests examined the associations between categorical variables, such as education versus occupation, occupation versus savings, and marital status versus savings (Habib and Jubb 2015). A significance threshold of 0.10 ($\alpha = 0.10$) was set to reflect a 90% confidence level, considering the exploratory nature of the research and the contextual limitations regarding sample diversity (Zhou et al. 2020). For continuous variables, Pearson correlation analysis evaluated the relationship between household income and expenditure. Additionally, an independent samples t-test compared the number of earning members among households with and without reported savings (Zhou et al. 2020). All statistical analyses were conducted using Python and validated for consistency and reliability.

2.4 Estimation of normalized difference vegetation index (NDVI)

The NDVI assesses the health of plants by analyzing their light reflectance at specific wavelengths, where certain wavelengths are absorbed and others are reflected. Due to its consideration of variations in lighting conditions, surface slope, exposure, and other environmental factors, the NDVI is selected for global vegetation monitoring (Maharjan et al. 2021; Rani et al. 2018; Sultana and Satyanarayana 2020). NDVI can be calculated through Eq. 1.

$$NDVI = \frac{NIR - RED}{NIR + RED} \tag{1}$$

For Landsat-8 band 5 is NIR and band 4 is RED.

Fig. 2 Flowchart of the research design

2.5 Estimation of normalized difference Built-up index (NDBI)

The built-up area is measured using the NDBI, which uses the range between short-wave infrared (SWIR) and near-infrared light (NIR). When the NDBI value ranges from −1 to 1, a negative value denotes a pixel's location in an undeveloped area, while a positive number denotes a concentration of developed areas (Maharjan et al. 2021). The NDBI computation is shown in Eq. 2.

$$NDBI = \frac{SWIR - NIR}{SWIR + NIR} \tag{2}$$

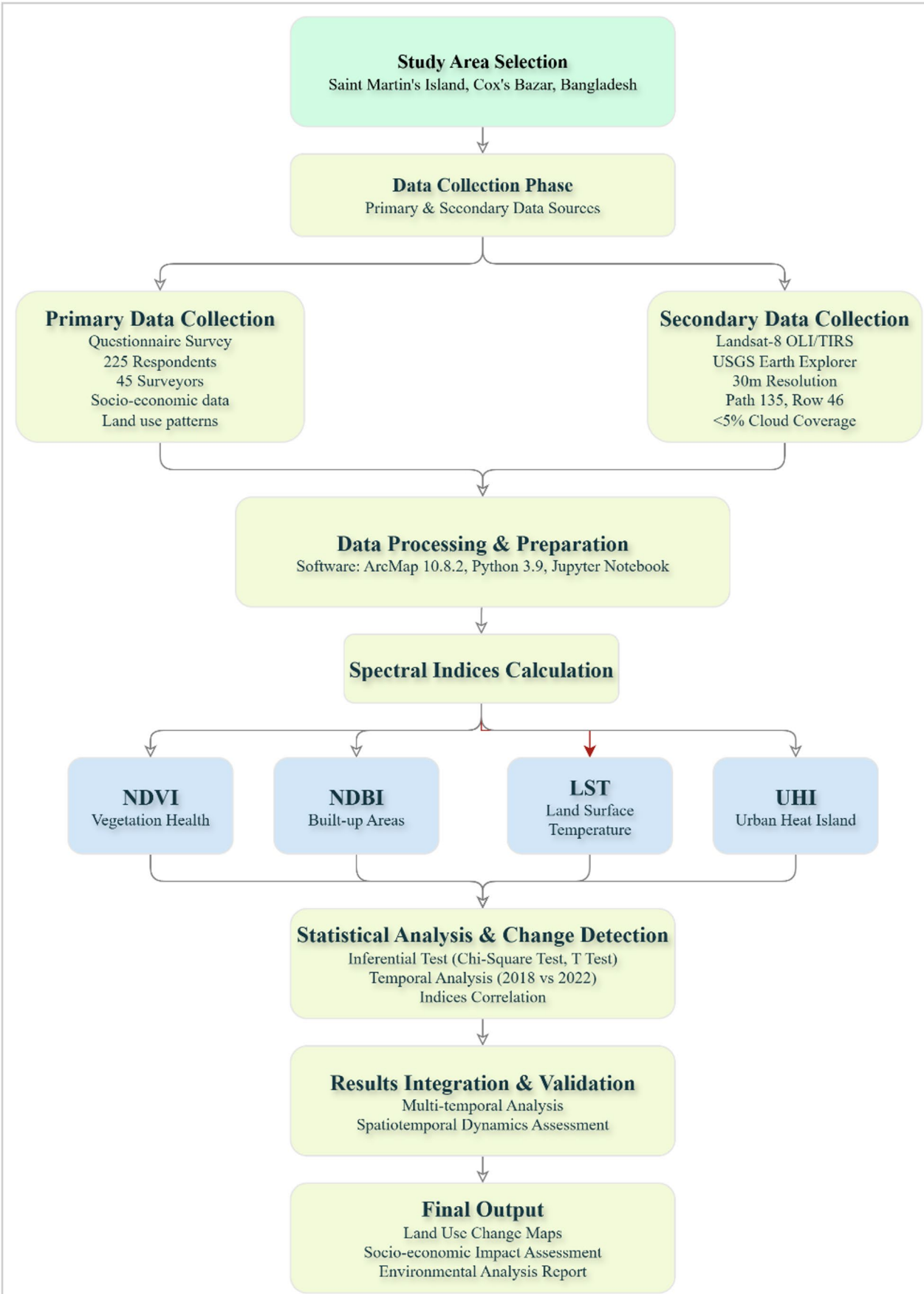
Here, SWIR is band 6 and NIR is band 5 for Landsat-8 data.

2.6 Estimation of land surface temperature (LST) and urban heat Island (UHI)

The LST influences the land surface physics by mediating the exchange of water and energy with the atmosphere. Various techniques are essential to derive LST from satellite thermal data, including sensor radiometric calibration, corrections for atmospheric and surface emissivity, and characterization of regional land cover variability, among others (Avdan and Jovanovska 2016). It is inappropriate to directly compare temperature values represented by LST at different times, due to modifications in near-surface atmospheric water vapor concentration caused by seasonal variations and inter-annual fluctuations of atmospheric conditions (Weng et al. 2004). Consequently, the primary focus of this study is on the UHI intensity and its spatial distribution within the research region. UHI intensity is defined as the temperature difference between the hottest parts of the urban area (LST) and the ambient non-urban surroundings (Chen et al. 2006). This effect can be assessed using individual thermal images and compared across multiple periods. However, it is imperative to compute LST prior to determining UHI intensity.

Thermal band 10 was used to collect the LST for Landsat-8, which can be accomplished in six steps (Avdan and

Content courtesy of Springer Nature, terms of use apply. Rights reserved.



Content courtesy of Springer Nature, terms of use apply. Rights reserved.

Jovanovska 2016; Gutman et al. 2013; He et al. 2019; Weng et al. 2004).

Step 1 calculates the radiance from band 10 (Eq. 3).

$$L_\gamma = M_L * QCAL + A_L \tag{3}$$

Where Lγ is spectral radiance, M_L is the band-specific multiplicative rescaling factor from the metadata, A_L is the band-specific additive rescaling factor from the metadata.

Step 2 is after converting the spectral radiance, which needs to be converted to atmospheric temperature brightness (Eq. 4).

$$TB = \frac{K2}{\ln(\frac{K1}{L_\gamma} + 1)} - 273.15 \tag{4}$$

TB is the atmospheric brightness temperature in Kelvin, and K2 and K1 are the band-specific thermal conversion constants, 1321.0789 and 774.8853, respectively.

Step 3 is to calculate NDVI, which is essential for calculating LST for Landsat-8 (Eq. 5).

$$NDVI = \frac{NIR - RED}{NIR + RED} \tag{5}$$

Where NIR is band 5 and Red is band 4.

Step 4 is to calculate the proportion of vegetation using maximum and minimum NDVI (Eq. 6).

$$P_v = \left(\frac{NDVI - NDVI_{min}}{NDVI_{max} - NDVI_{min}}\right)^2 \tag{6}$$

Step 5 is to calculate Land surface emissivity using Pv (Eq. 7).

$$LSE = 0.004 * P_v + 0.986 \tag{7}$$

Step 6 is to calculate the LST in degrees Celsius (Eq. 8),

$$T = TB/\left[1 + \left(\gamma * \frac{TB}{c2}\right)\ln(LSE)\right] \tag{8}$$

Where γ is the wavelength of emitted radiance, c2 is h*c/s = 1.4388*10-2mK = 14388μmK, where h is Plank's constant, which is 6.626*10–34 Js, s is Boltzmann constant, which is 1.38*10–23 J/K, and c is the velocity of Light, which is 2.994*10^3 m/s.

UHI is often described as the difference between rural and urban areas (Hu et al. 2019). Quantification can also define this by considering the' local and regional climate change of urban surfaces. For calculating the UHI, the LST value was used (Eq. 9).

$$UHI = \frac{T - T_m}{T_{sd}} \tag{9}$$

where T is LST, Tm is LST mean, and Tsd is the standard deviation of LST.

The UHI intensity change between 2018 and 2022 was quantified through pixel-by-pixel difference analysis of the UHI maps. The UHI change (ΔUHI) was calculated using Eq. 10:

$$\Delta UHI = UHI_{2022} - UHI_{2018} \tag{10}$$

Where UHI_{2022} and UHI_{2018} represent the standardized UHI values for each corresponding pixel in 2022 and 2018, respectively

3 Results

3.1 Socio-economic analysis

The demographic analysis indicates that large household sizes and limited earning capacity characterize Saint Martin's Island. As detailed in Tables 3 and 70.22% of households consist of 5 to 10 members, while 24.89% have fewer than five members. Notably, 88% of households have fewer than three earners, with only 10.67% employing three to five earners, and a mere 1.33% having more than five earners. The occupational structure predominantly features business activities, with 46.22% of respondents engaged in commerce, followed by day laborers (15.11%) and farmers (10.22%) (Table 4). Fishermen and housewives

Table 3 Distribution of total and earning family members in the household (*N*=225)

Family Size	Percentage	Earning Members	Percentage
Below 5	24.89	Below 3	88.00
5–10	70.22	3–5	10.67
Above 10	4.89	Above 5	1.33
Total	100	Total	100

Source: Field Survey, 2023

Table 4 Occupation of the respondents (*N*=225)

Occupation	Percentage	Occupation	Percentage
Businessman	46.22	Service	3.11
Day Laborer	15.11	Retired	2.22
Farmer	10.22	Unemployed	1.78
Fisherman	7.11	Others	7.11
Housewife	7.11	Total	100

Source: Field Survey, 2023

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

each constitute 7.11% of the sample, whereas employment in the service sector accounts for only 3.11% of respondents. Educational attainment remains limited among the population, with a substantial proportion attaining only primary education and a significant number being illiterate individuals (Fig. 3a). The income distribution reveals that 35% of respondents earn between 10,001 and 20,000 BDT monthly, while 28% earn between 20,001 and 30,000 BDT (Fig. 3b). This suggests that most households (63%) have monthly incomes ranging from 10,000 to 30,000 BDT, primarily generated from tourism and business activities. Monthly household expenditure patterns demonstrate that 43% of respondents allocate between 10,000 and 20,000 BDT (Fig. 3c). At the same time, savings capacity remains severely Limited, with 61% of households maintaining savings below 5,000 BDT and 19% saving between 5,000 and 10,000 BDT (Fig. 3d).

The chi-square test demonstrated a significant association between educational level and occupation (χ^2 = 110.96, $p<0.001$), indicating that educational attainment has a substantial influence on occupational engagement within the island community. Table 5 shows the details of

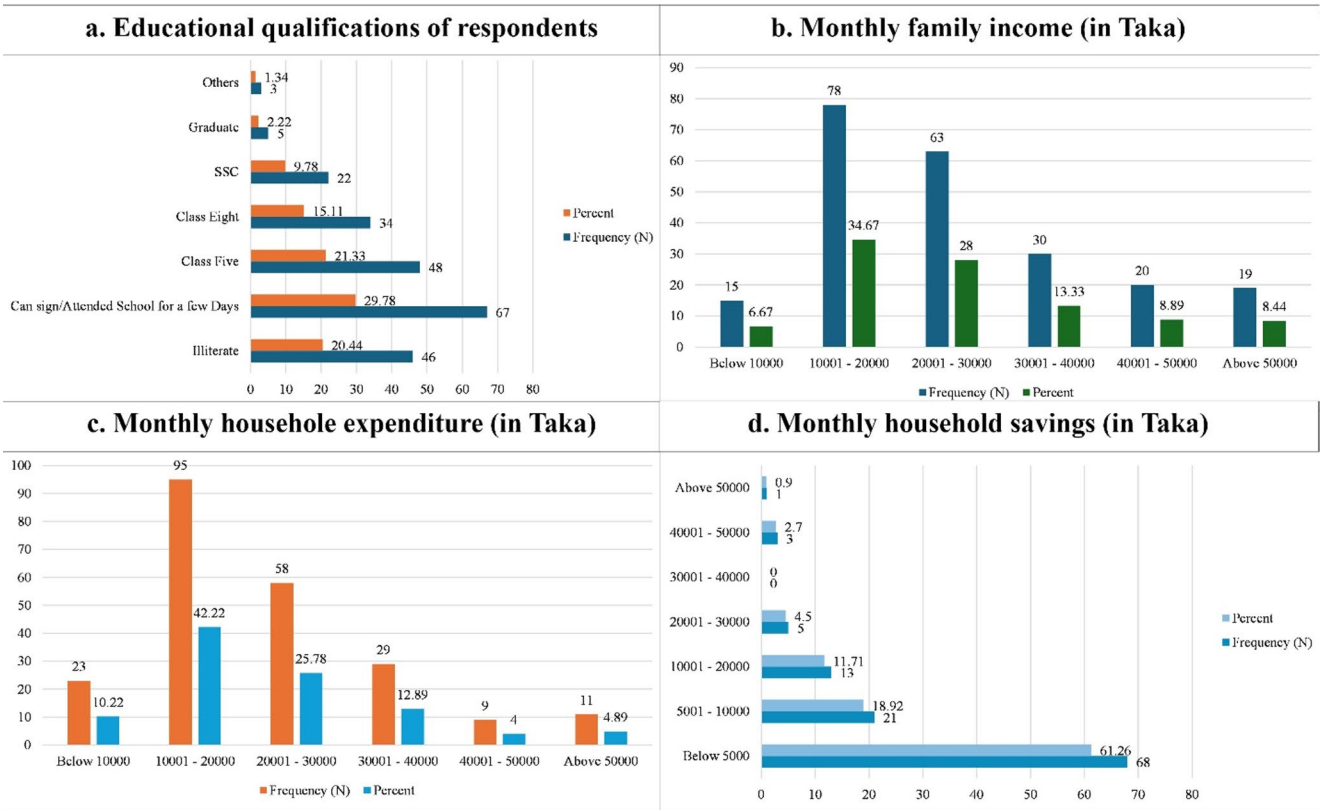


Fig. 3 Socio-economic status in Saint Martin's Island, Bangladesh: (**a**) Educational qualification; (**b**) Monthly family income; (**c**) Monthly household expenditure; (**d**) Monthly savings

Table 5 Summary of Inferential statistical tests at 90% confidence level

Test Type	Variables Involved	Test Statistic	*p*-value	Result (90% CI)	Interpretation
Chi-square	Education × Occupation	110.96	0.0000	Significant	Education level influences occupation type
Chi-square	Occupation × Savings (Yes/No)	9.02	0.0605	Significant	Occupation affects saving behaviour.
Chi-square	Marital Status × Savings	17.95	0.3919	Not significant	No clear impact of marital status on savings
Pearson Correlation	Income × Expenditure	$r=0.857$	0.0000	Significant	Strong positive relationship
T-test	Earning Members × Savings	t=0.285	0.7759	Not significant	Number of earners does not affect savings

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

the tests. Furthermore, a significant relationship was identified between occupation and savings status (χ^2 = 9.02, p=0.0605), suggesting that the type of employment may impact a household's ability to save. Conversely, no significant correlation was observed between marital status and savings (p=0.3919), implying that marital status does not serve as a primary determinant of saving behavior in this context. A strong positive correlation (r=0.857, p<0.001) was identified between monthly income and expenditure, indicating that households with higher earnings tend to incur higher expenses. This reflects a consumption-driven economic framework where income levels closely correspond to expenditure. An independent T-test comparing households with and without savings revealed no statistically significant difference regarding the number of income earners (t=0.285, p=0.7759), suggesting that the number of income earners does not necessarily influence a household's capacity to save.

3.2 Land use changes analysis

3.2.1 Pattern of land use changes analysis through primary data

The examination of land use change patterns reveals three predominant transformation trajectories over the past five years: the conversion of bare land to commercial use, agricultural land to commercial use, and residential areas to commercial use (Table 6). The primary catalysts for these transformations include opportunities for economic development, enhanced profitability within the business sector, benefits derived from the tourism industry, resort construction activities, and increased soil salinity, which impacts agricultural productivity. Current land utilization patterns indicate that 53% of the land is dedicated to agricultural use, constituting the largest single category, followed by bare land areas (Fig. 4c). An analysis of the driving factors behind these changes demonstrates that business-related factors account for over 80% of land use conversions, with the development of the tourism industry (41.28%) and resort construction (20.49%) serving as the most significant catalysts (Fig. 4d). The fragmentation of land holdings is evident, with larger plots being subdivided into smaller parcels to accommodate diverse business activities (Fig. 4a, b). This process is driven by multiple factors, including economic pressures, tourism infrastructure requirements, and changing livelihood strategies.

Table 6 Patterns and causes of land use change in the last five years

Patterns of Land Uses	Causes
Bare land to Business	Better economic development, better business benefits, benefits from tourism, and building new resorts
Agriculture to Business	Better economic development, more business benefits, benefits from tourism, building new resorts, increasing salinity
Residential to Business	Economic development, increasing need for resorts

Source: Field Survey, 2023

3.2.2 NDVI and NDBI analysis from the perspective of land use changes

The NDVI analysis reveals significant changes in vegetation between 2018 and 2022. The northern region, particularly the central section, experienced notable declines in vegetation during this period (Fig. 5). NDVI values in the northern area showed a marked reduction in 2022 compared to 2018, indicating a substantial loss of vegetation cover. Conversely, the southern region maintained relatively higher NDVI values in 2022, suggesting better preservation of vegetation cover.

The NDBI analysis demonstrates corresponding increases in built-up areas, with the most pronounced changes occurring in the northern section of the island between 2018 and 2022 (Fig. 6). The spatial distribution of increased NDBI directly correlates with areas experiencing the most intensive tourism-related development.

3.2.3 LST and UHI analysis from the perspective of land use changes

LST analysis for December 2018 and 2022 reveals significant changes in the thermal environment. The northern region experienced temperature increases of up to 0.2 °C during this period (Fig. 7). This temperature elevation corresponds spatially with areas showing increased NDBI values and decreased NDVI values, indicating a direct relationship between the expansion of built-up areas and an increase in surface temperature.

UHI analysis demonstrates more significant thermal impacts (Fig. 8). In 2018, the maximum UHI intensity reached 2.6 °C, while in 2022, this value increased to 3.8 °C. The difference analysis reveals that the northwestern region experienced the most dramatic UHI intensification, with temperature increases reaching 1.9 °C. The spatial pattern of UHI intensification closely aligns with areas of concentrated tourism infrastructure development.

3.2.4 Correlation analysis of remote sensing indices

Comprehensive correlation analyses were performed to quantify the relationships between land use change indicators (NDVI, NDBI) and thermal environment parameters

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

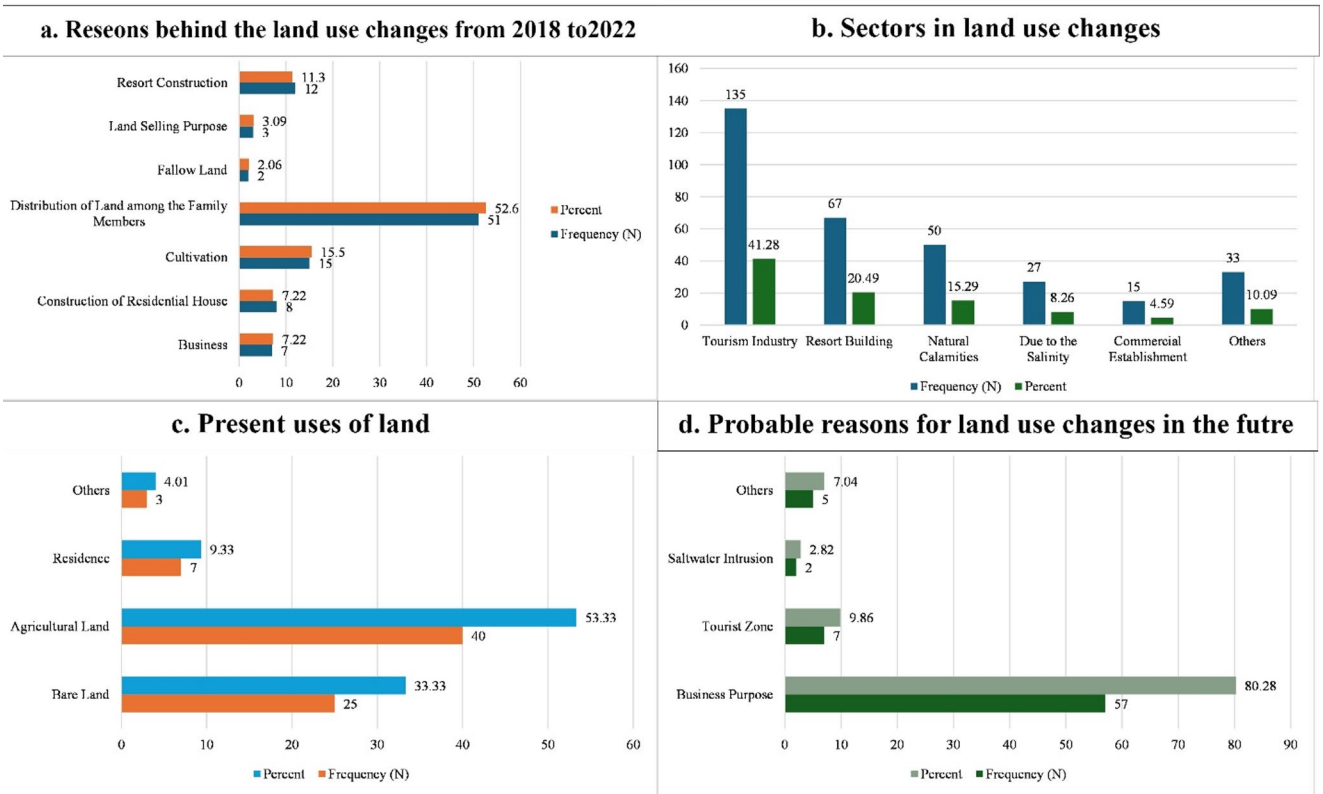


Fig. 4 Pattern and causes of land-use changes in Saint Martin's Island, Bangladesh: (**a**) Causes of land use change from 2018 to 2022; (**b**) Sector in land use change; (**c**) Present land use pattern; (**d**) Probable reasons for land use changes in the future. Source: Field Survey, 2023

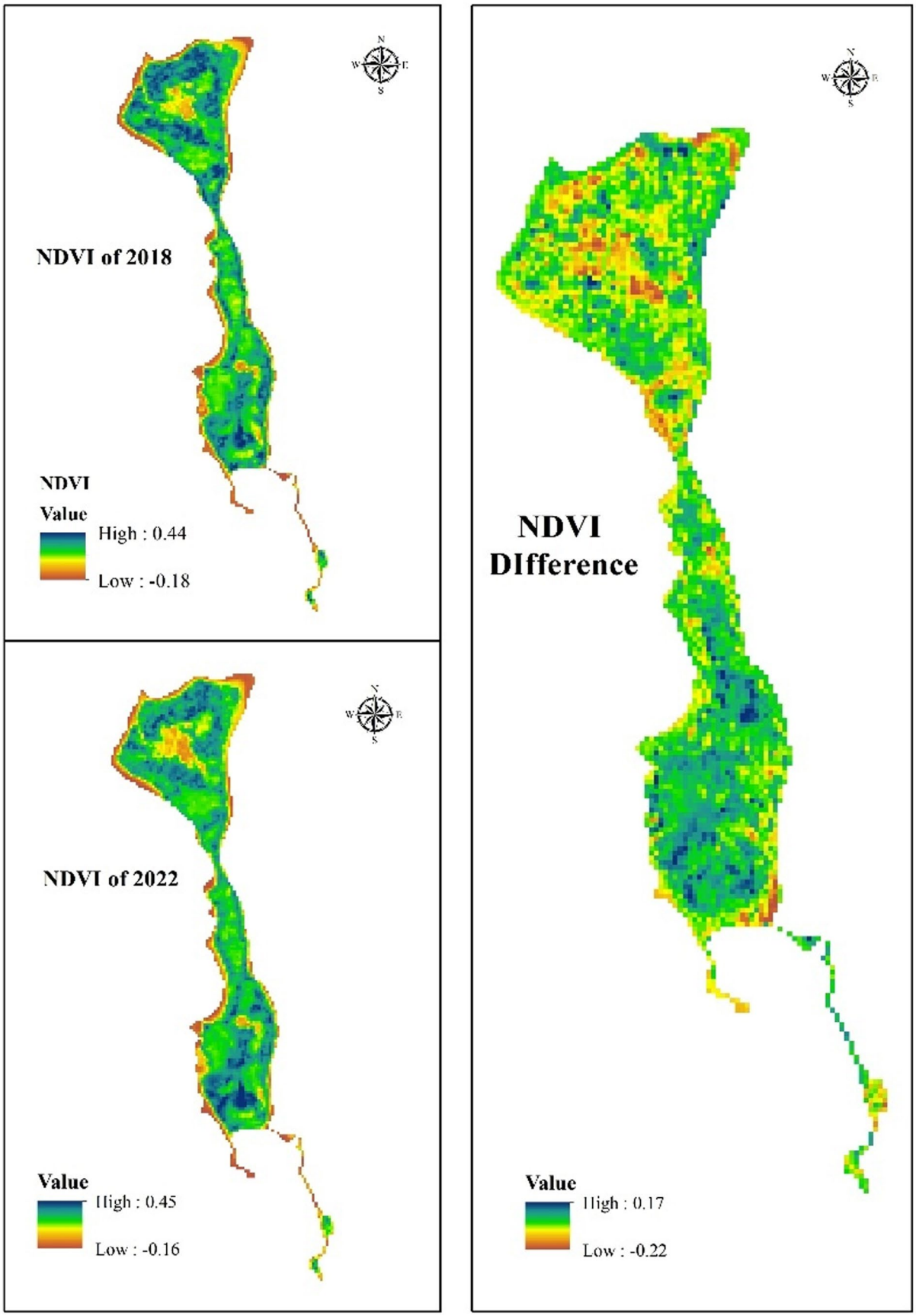
(LST, UHI), employing both Pearson and Spearman correlation coefficients (Fig. 9). The analysis identified strong negative correlations between vegetation cover and thermal parameters. NDVI exhibited moderate negative correlations with both LST (Pearson: −0.576, Spearman: −0.697) and UHI (Pearson: −0.350, Spearman: −0.144), with p value less than <0.05, suggesting that areas with higher vegetation cover experienced lower surface temperatures. Conversely, NDBI demonstrated positive correlations with thermal parameters (LST: Pearson's $r=0.393$, Spearman's $\rho=0.444$; UHI: Pearson's $r=0.393$, Spearman's $\rho=0.444$), confirming that built-up areas are associated with elevated surface temperatures. The correlation patterns intensified by 2022, with NDVI-UHI correlations increasing to Pearson's $r = -0.474$ and Spearman's $\rho = -0.462$, while NDVI-LST correlations remained significant (Pearson's $r = -0.474$, Spearman's $\rho = -0.612$). NDBI correlations with thermal parameters persisted as positive relationships (LST: Pearson's $r=0.311$, Spearman's $r=0.324$; UHI: Pearson's $r=0.311$, Spearman's $r=0.324$), although their magnitude was reduced compared to 2018, potentially indicating spatial saturation effects of urbanization.

These correlation patterns substantiate the anticipated inverse relationship between vegetation cover and surface temperature, while also confirming that expansion of built-up areas directly contributes to the deterioration of the thermal environment. The strengthening of NDVI-UHI correlations from 2018 to 2022 suggests that vegetation loss has become an increasingly critical factor influencing the formation of local heat island phenomena.

3.2.5 Analysis of the impact of land use changes on LST and UHI

The assessment of land use change impacts reveals contrasting community perspectives. Positive impacts are primarily associated with economic benefits, with economic development and employment opportunities each accounting for over 30% of responses (Fig. 10a). These economic benefits are directly Linked to the growth of the tourism sector and related business opportunities. Negative impacts are primarily driven by environmental concerns, with environmental pollution accounting for 28.90% of responses, followed by concerns about land loss at 24.31% (Fig. 10b). These community perceptions align with the quantitative environmental changes documented through remote sensing analysis, particularly the observed increases in surface temperature and UHI effects. The convergence of quantitative remote

Content courtesy of Springer Nature, terms of use apply. Rights reserved.



Content courtesy of Springer Nature, terms of use apply. Rights reserved.

Fig. 5 NDVI of Saint Martin's Island of 2018, 2022, and the difference between 2018 and 2022

sensing data with community perceptions validates both the environmental changes occurring on the island and the local population's awareness of these transformations. The documented decreases in NDVI, increases in NDBI, and elevation of LST and UHI values provide empirical support for community concerns regarding environmental degradation.

4 Discussion

4.1 Linkages between Socio-economic conditions and land use transformations

The findings of this study demonstrate a complex interaction between tourism-oriented economic opportunities and land use modifications on Saint Martin's Island. The socio-economic analysis indicates that, despite the prevalence of large families (70.22% comprising 5–10 members) and a limited number of earning members (88% having fewer than three earners), residents have progressively shifted towards tourism-related enterprises (46.22% engaged in business activities). This transition constitutes a rational economic adaptation to the expanding tourism sector, which attracts approximately 8,000 visitors daily during peak seasons (Ahmed et al. 2023). The income levels reported, ranging from 10,001 to 30,000 BDT, although higher than historical averages for the island, remain below the standards observed in national urban centers (Ara et al. 2021). This economic pressure, combined with the availability of tourism opportunities, has become a significant driver of land use conversion. The identified pattern of land conversion transforming bare land into commercial spaces (primarily for resort development), converting agricultural land to commercial use, and converting residential areas into commercial zones directly mirrors the economic drivers documented within the socio-economic analysis.

The inferential statistical findings offer deeper insights beyond descriptive patterns. The significant association between education and occupation highlights the role of education in shaping economic participation. Respondents with higher education levels were more likely to engage in business or skilled jobs, while those with limited education often worked in labor-intensive or informal sectors.

The association between occupation and savings suggests that households engaged in business or consistent income-generating activities tend to save more than those relying on irregular or subsistence employment (Kofarmata and Danlami 2021; Zhou et al. 2020). Nevertheless, the absence of a correlation between marital status and savings challenges conventional assumptions that married individuals save more, possibly due to substantial household expenditure or cultural spending customs on family needs (Chikeya and Ntsalaze 2025). Moreover, the strong correlation between income and expenditure suggests a limited financial surplus among households, as increased income is often accompanied by proportionally higher spending, thereby constraining long-term savings and investments (Sodokin 2023). Contrary to common beliefs, the number of members earning does not significantly influence a household's capacity to save, potentially owing to low individual earnings or a high number of dependents (Ndamsa et al. 2023). These inferential insights underscore the importance of targeted interventions, such as education-focused skill development and financial literacy programs, in enhancing income diversification and savings capacity within vulnerable island communities.

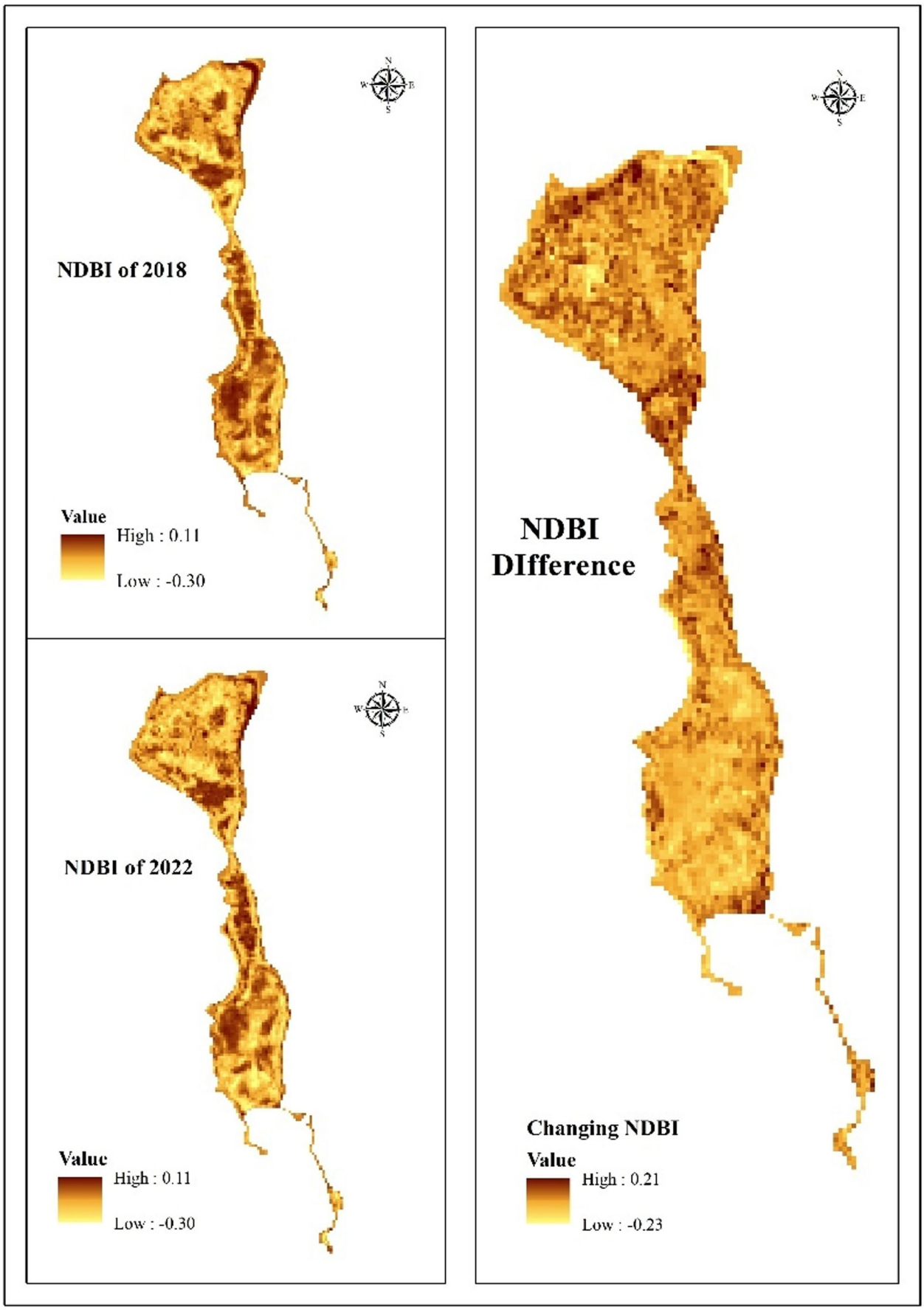
4.2 Remote sensing evidence of Tourism-Induced environmental changes

The integration of NDVI and NDBI analyses provides quantitative evidence supporting the socio-economic drivers of land use change. The decline in NDVI values in the northern region between 2018 and 2022, particularly in the central section, directly correlates with the areas experiencing the highest tourism development pressure. This finding aligns with global patterns observed in tourism-intensive small islands, where vegetation loss typically accompanies infrastructure development (Priskin 2001; Yadav 2024). Similarly, Chu observed that tourism-related locations on Hainan Island exhibited markedly lower NDVI values and experienced a 72% expansion in urban areas relative to sites not associated with tourism (Chu et al. 2020). The corresponding increase in NDBI values in the same northern region provides compelling evidence of the expansion of built-up areas reported by respondents. LST exhibits a strong correlation with negative NDVI and positive NDBI, which is consistent with our findings, where areas with increased NDBI (such as tourism infrastructure) display elevated LSTs. This relationship has been documented in urban studies globally, where the replacement of vegetation with built surfaces fundamentally alters the thermal properties of the landscape (Dong et al. 2022; Guha et al. 2018).

4.3 Thermal environment impacts and urban heat Island (UHI) formation

The LST and UHI analyses reveal concerning changes in the thermal environment that directly Link to the documented land use transformations. The temperature increase of up to 0.2 °C in LST and the UHI spike from 2.6 °C to 3.8 °C between 2018 and 2022 represent significant environmental

Content courtesy of Springer Nature, terms of use apply. Rights reserved.



Content courtesy of Springer Nature, terms of use apply. Rights reserved.

Fig. 6 NDBI of Saint Martin's Island of 2018, 2022, and the difference between 2018 and 2022

changes for a small island ecosystem. The northwestern region showing the most dramatic UHI increase (up to 1.9 °C) corresponds precisely with the areas identified by respondents as experiencing the most intensive resort development. The nearest BMD station data were utilized to validate the LST and UHI. December temperature records show maximum temperatures of 28.37 °C (2018) and 30.21 °C (2022), alongside minimum temperatures of 16.58 °C (2018) and 18.13 °C (2022). The observed increase in maximum temperatures (1.84 °C and minimum temperatures (1.55 °C at the regional level provides supporting context for the thermal variations identified through satellite-derived LST analysis. These findings are in Line with other studies linking urbanization to higher surface temperatures. For example, Chu et al. observed that the median LST at tourism sites in Hainan was about 1.2 °C higher than at non-tourist sites from 2000 to 2019 (Chu et al. 2020). Similarly, one study reported that South Asian city centers with dense built-up cover exhibited strong heat island characteristics (Maharjan et al. 2021). The direct relationship observed between the expansion of built-up areas and increased LST on Saint Martin's Island reflects the well-documented dynamics in Bangladesh's megacities. For example, a study on Dhaka found that from 1990 to 2020, the urban area increased by 40% while green space declined by 41% (Islam et al. (2023), and another study directly linked the density of built-up areas in Dhaka with higher surface temperatures and more intense heat waves (Islam et al. 2024a, 2024b). This study's findings suggest that this same urbanization-driven warming process is now actively occurring on a sensitive coastal island, driven by tourism rather than industrialization.

The regions exhibiting the most significant development correlate with elevated LST and UHI effects, thereby substantiating the causal mechanism: the replacement of

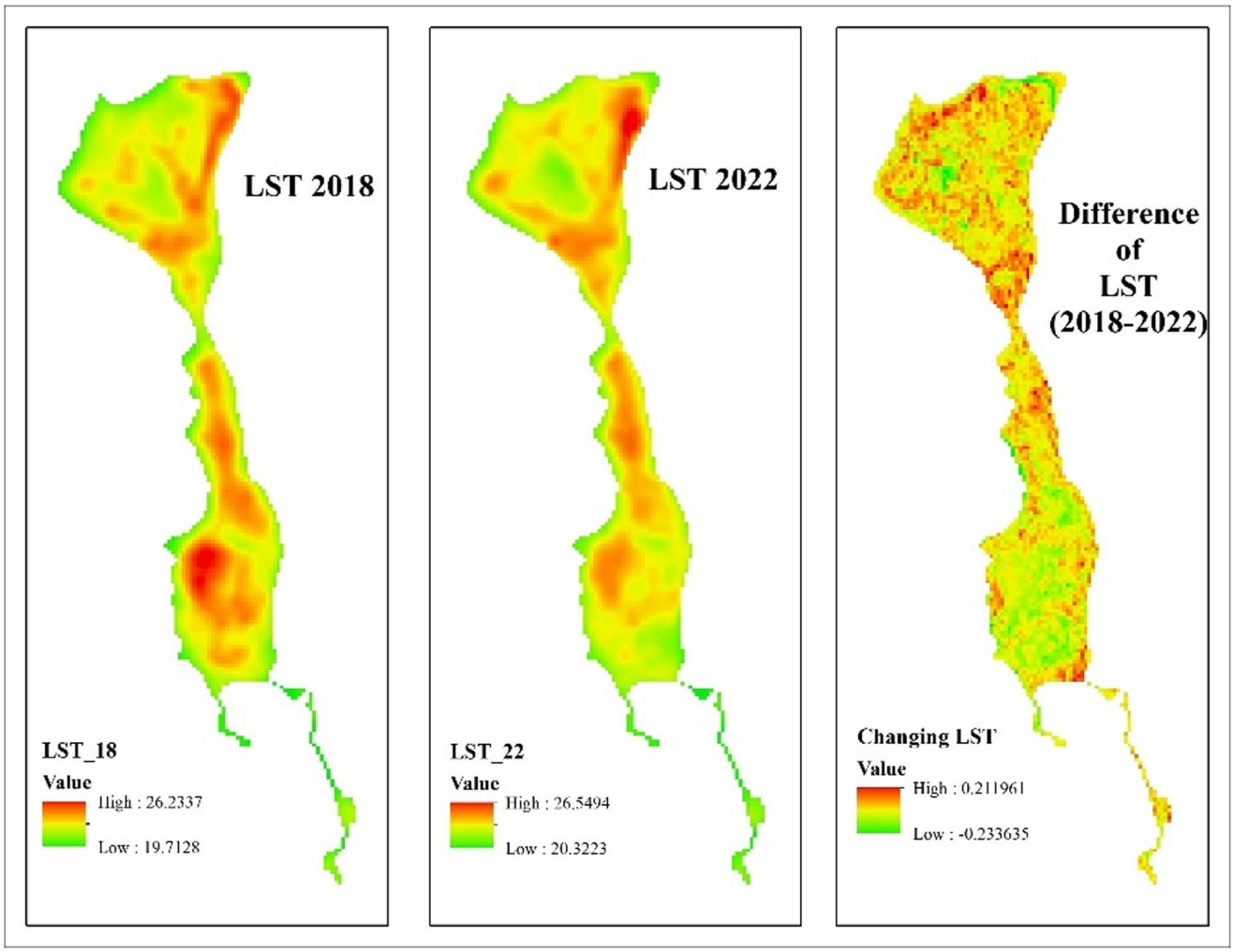


Fig. 7 LST of Saint Martin's Island of 2018, 2022, and the difference between 2018 and 2022

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

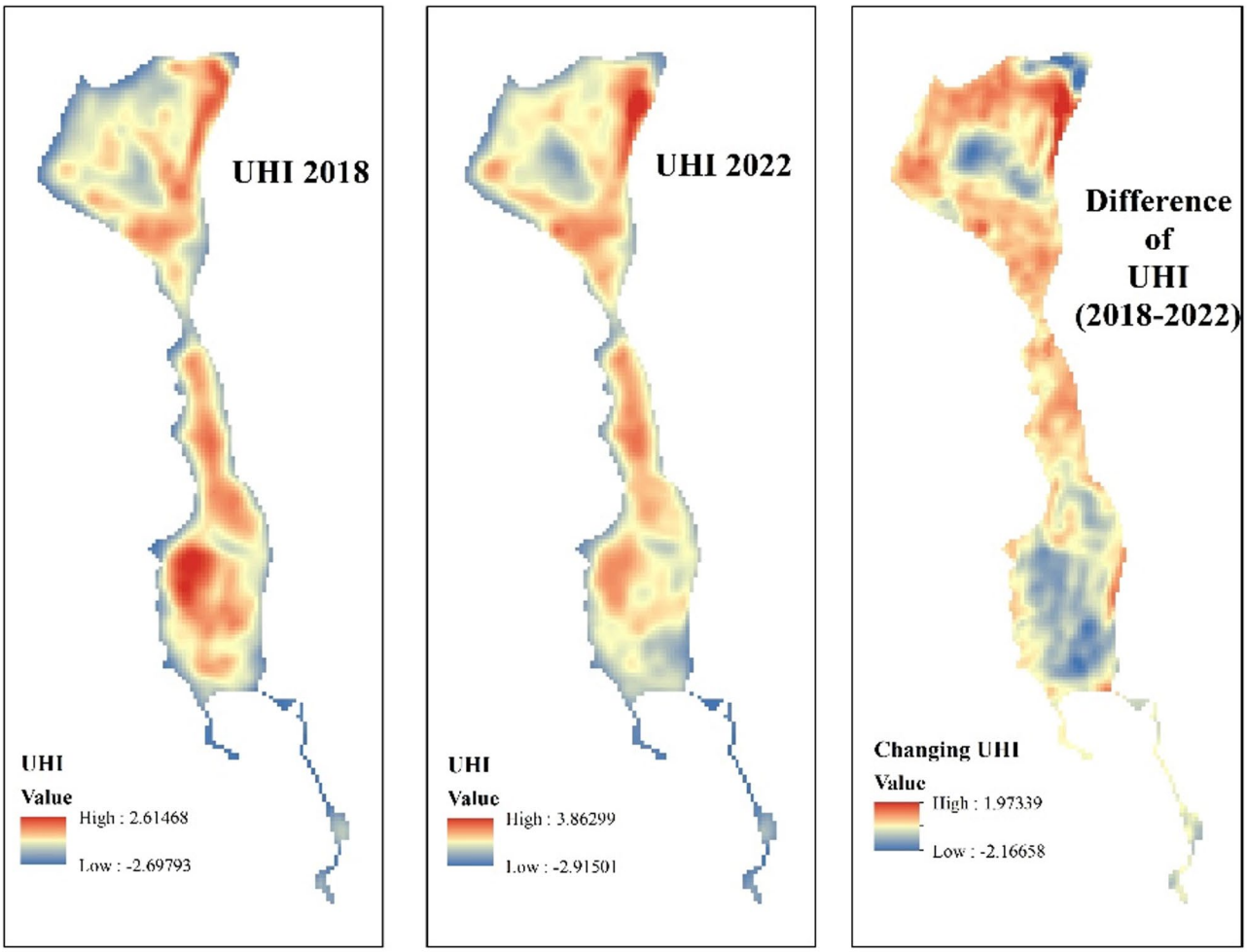


Fig. 8 UHI of Saint Martin's Island of 2018, 2022, and the difference between 2018 and 2022

vegetation with impervious surfaces leads to increased heat retention. This mechanism is well documented, as vegetation facilitates cooling through evapotranspiration, whereas concrete and asphalt trap and reradiate heat (Cetin et al. 2024; Safarrad et al. 2021). The close interrelation is evident in the correlation analyses, which reveal a significant negative correlation between UHI and NDVI, as well as a positive correlation with the built-up index (Cetin et al. 2024). This study observed a consistent pattern at St. Martin's: regions characterized by lower NDVI and higher NDBI values correspond to the hottest areas in the northwest, suggesting that the observed warming is directly linked to land-use changes driven by tourism activities. A comparison of these findings with existing literature offers additional clarification. One study also reported that, across various land uses, LST and NDVI were inversely related. In contrast, built-up indices demonstrated a positive correlation with increased urbanization temperatures (Safarrad et al. 2021), and emphasize our findings that declining vegetation (NDVI) and expanding infrastructure (NDBI) contribute to increased LST and UHI. Notably, Studies also showed that tourism zones planted with green space had lower LST than urban areas (Cameron et al. 2020; Safarrad et al. 2021). The results of this study show that greener areas, such as the unbuilt south, tend to stay cooler. Tourism-driven land conversion causes vegetation loss and expansion, leading to increased LST/UHI.

4.4 Sustainable development challenges

This study on Saint Martin's Island highlights the conflicts between economic development and environmental preservation that are prevalent in small island tourism. While tourism enhances the economy and creates employment opportunities, it concurrently increases environmental costs, as evidenced by community perceptions and remote sensing

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

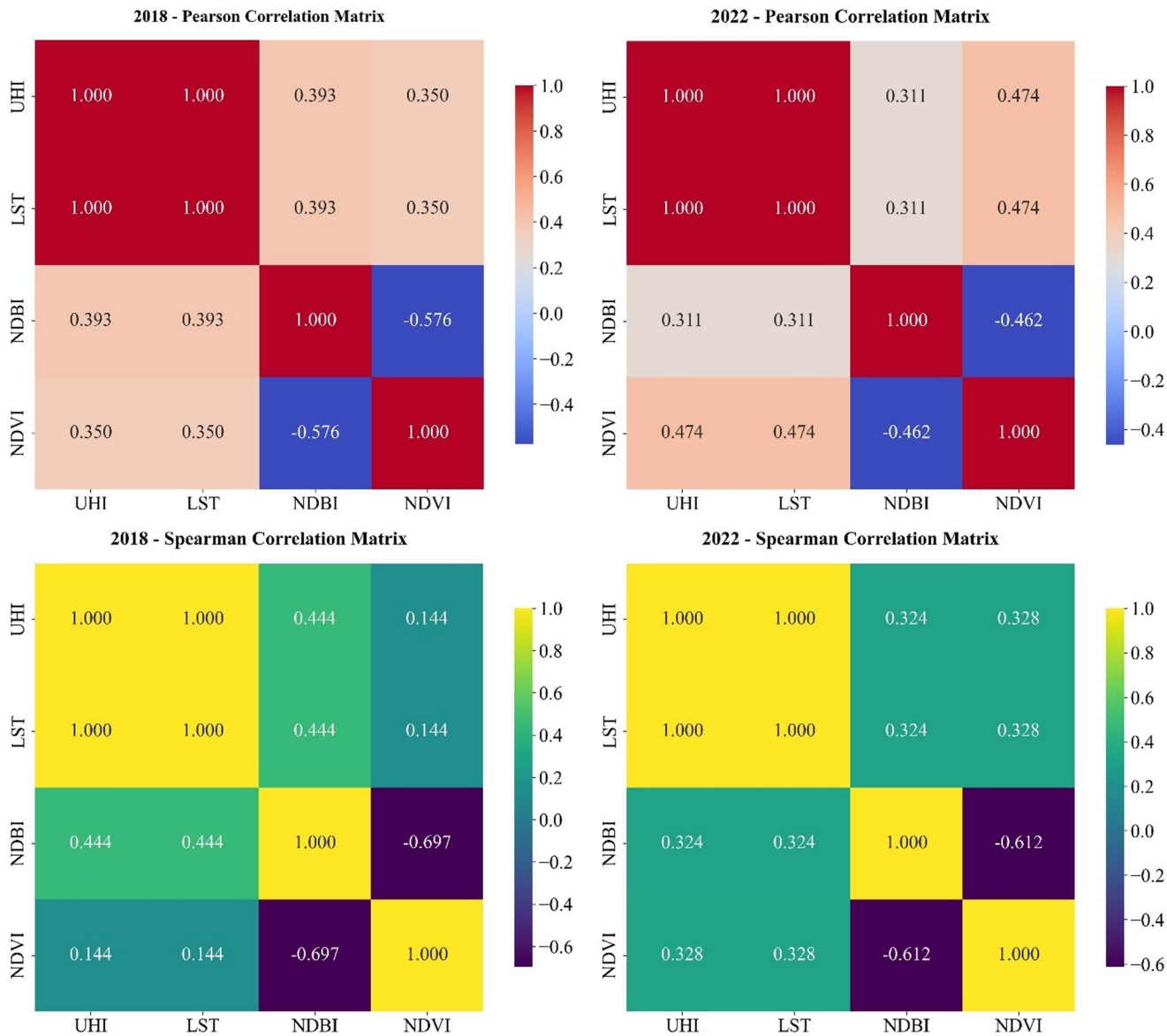


Fig. 9 Correlation analysis between NDVI, NDBI, LST, and UHI

data indicating substantial alterations in NDVI, NDBI, LST, and UHI within just four years. The encroachment of agricultural land by development activities and rising salinity levels poses threats to food security, increasing dependence on external resources. This pattern mirrors global trends where tourism expansion jeopardizes traditional livelihoods (Scheyvens and Momsen 2008; Sharpley 2012). Islands remain susceptible to biodiversity loss and the impacts of climate change, and the observed environmental transformations suggest that current development practices may be unsustainable unless complemented by adaptive strategies that reconcile economic growth with environmental conservation.

4.5 Policy implications

Based on the study's findings and relevant literature, immediate implementation of evidence-based policy interventions is essential to prevent irreversible ecological collapse. First, establish mandatory Environmental Impact Assessments (EIAs) for all tourism infrastructure projects, with specific thermal impact thresholds that trigger mitigation requirements (Skiniti et al. 2024). Second, implement a tourism carrying capacity framework that limits daily visitor numbers to a minimum, with dynamic adjustments based on real-time environmental monitoring (Marsiglio 2017). Third, mandate green building standards that require proper

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

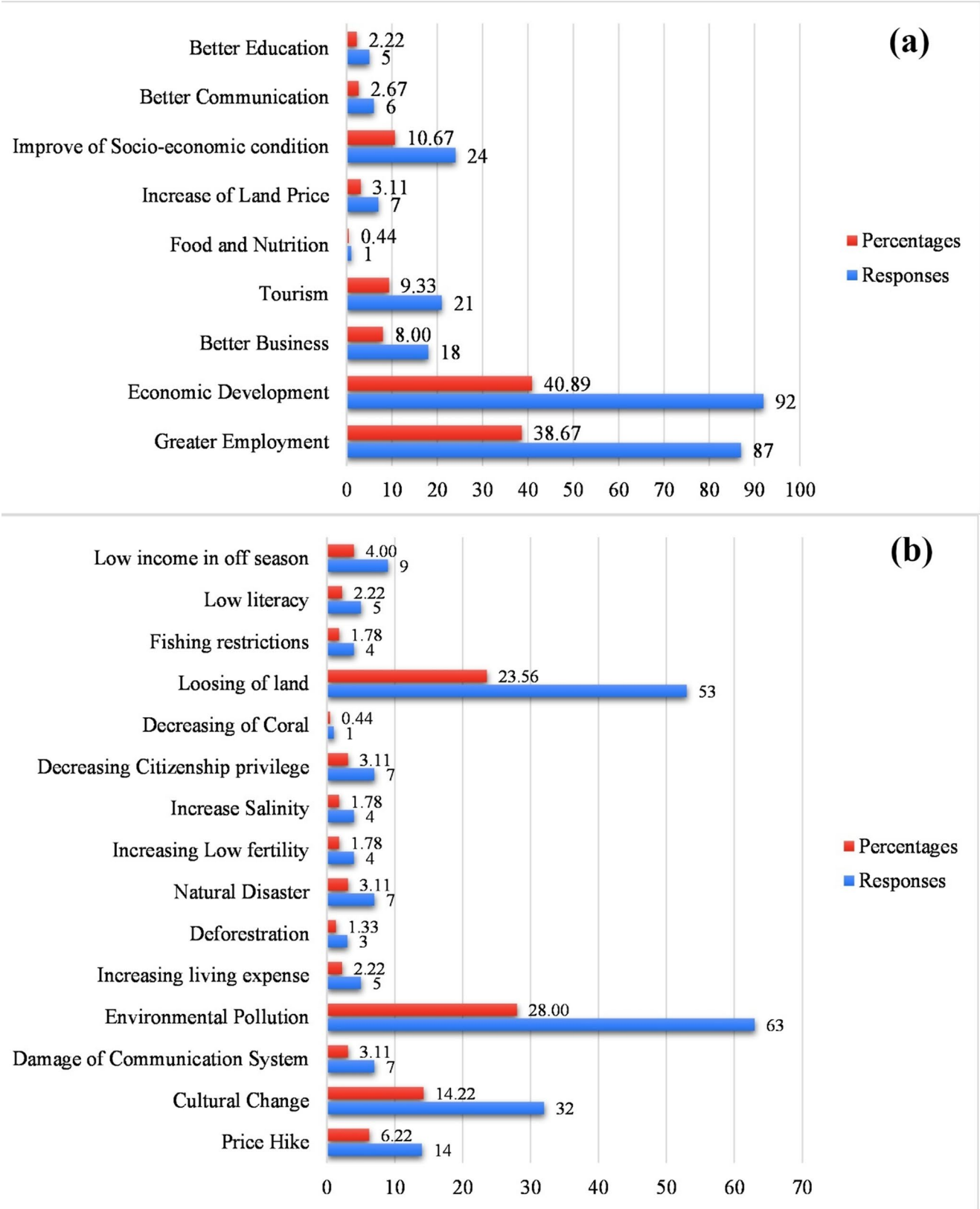


Fig. 10 Positive and negative impacts of land use changes: (**a**) Positive impacts; (**b**) Negative impacts. (Source: Field Survey, 2023)

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

vegetation coverage for all new resort developments and retrofit existing structures with cooling technologies, such as green roofs and permeable surfaces (Shafique et al. 2018). Fourth, create a spatial zoning system that designates the southern region as a conservation zone, while concentrating controlled development in already-impacted northern areas to prevent further spatial expansion of thermal hotspots (Li et al. 2023). Fifth, establish a tourism revenue-sharing mechanism allocating a specific amount of tourism income to ecosystem restoration programs, including reforestation initiatives and coral reef conservation (Zhang et al. 2024).

4.6 Limitations and future directions

This study recognizes several Limitations that merit consideration. The sample size, consisting of 225 households, was representative of but limited by logistical challenges encountered during the peak tourism season, a period when residents were engaged in tourism-related activities. The data collection was conducted over two days; although intensive, this timeframe provides only a snapshot that may not fully encompass seasonal fluctuations in socio-economic conditions. Furthermore, the socio-economic data were not spatially linked, as the household surveys were not GPS-coordinated, which limits the ability to analyze spatial patterns or correlations with environmental variables. Also, the absence of ground-based meteorological stations on the island necessitated validation using data from the nearest station in Teknaf, which may introduce spatial variability in thermal measurements. Future research should incorporate prolonged monitoring systems, expand household surveys to include off-peak seasons, and undertake direct ground-truthing of thermal data to improve the accuracy of analysis.

5 Conclusion

This comprehensive study reveals critical linkages between socioeconomic conditions and land-use transformations on Saint Martin's Island. The demographic analysis shows that most large families (70.22%, comprising 5–10 members) face severe economic constraints, with limited earning capacity (88% having fewer than three income-generating members), creating substantial pressure for land conversion to tourism-related enterprises. The predominance of business activities, as indicated by 46.22% of respondents, reflects a rational economic adaptation to tourism opportunities; yet, this transition fundamentally alters traditional land-use patterns. Rising tourist arrivals have driven the construction of new resorts and hotels, resulting in the systematic conversion of agricultural land (affected by increasing salinity), bare land, and residential areas to commercial infrastructure. This economic imperative, while providing short-term financial benefits with average monthly incomes of 15,000 BDT, has precipitated irreversible changes in the island's spatial organization and ecological integrity.

The geospatial analysis, utilizing Landsat 8 data, further corroborates the significant environmental implications. NDVI values have exhibited a decline across urbanized zones, signifying a diminution in vegetation, whereas NDBI values have shown an increase, indicative of urban expansion. This trend coincides with a measurable rise in LST, with some hotspots experiencing a warming of up to 0.2 °C. The UHI effect intensified significantly, with peak UHI values increasing from 2.6 °C in 2018 to 3.8 °C in 2022. Correlation analysis implies that the reduction in vegetative cover, coupled with the proliferation of impervious surfaces, is a primary factor driving this thermal variation. Collectively, these findings suggest that tourism-driven land use modifications are not only reshaping the socio-economic fabric of the island but also augmenting its environmental vulnerabilities.

Immediate, evidence-based policy interventions are crucial to prevent irreversible ecological collapse. Key actions include implementing mandatory Environmental Impact Assessments for all new tourism projects, establishing a dynamic tourism carrying capacity to limit visitor numbers, and mandating green building standards to reduce thermal impact. A spatial zoning system will also help to protect the southern region, while tourism revenue can be reallocated to fund ecosystem restoration and conservation programs. Ultimately, future research should expand its temporal scope and incorporate biodiversity and hydrological parameters to provide a more comprehensive picture of ecological stress. While this study offers a robust integrative analysis, limitations include seasonal gaps in satellite data and limited temporal survey rounds. Nevertheless, the insights provide strong justification for adaptive land-use policies and long-term planning to ensure that Saint Martin's Island remains both habitable and economically vibrant.

Acknowledgements The authors thankfully acknowledge the people of Saint Martin's Island, Bangladesh, for providing primary data through a questionnaire survey for this research. The authors also thank the USGS for providing the dataset used in this study.

Author contributions **Md. Rakib Hasan Rony:** Writing – original draft. **Farhana Akter Jannat:** Investigation isualization. **Avishek Adhikari:** Investigation isualization. **Shahorare Ahomed:** Investigation isualization. **Md. Rabiul Haque:** Methodology critically for important intellectual content. **Md. Zakiur Rahman:** Methodology critically for important intellectual content. **Md. Mostafizur Rahman:** Conceptualization administration review and editing, design, approval of the version to be submitted.

Funding The authors declare that this research had no financial support or grants.

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

Data availability No datasets were generated or analysed during the current study.

Declaration

Declaration of generative AI The authors state that there is no use of generative AI in scientific writing.

Clinical Trial Number Clinical trial number is not applicable.

Competing interests The authors declare no competing interests.

References

AbdelRahman MAE (2023) An overview of land degradation, desertification and sustainable land management using GIS and remote sensing applications. Rend Lincei Sci Fis Nat 34(3):767–808. https://doi.org/10.1007/s12210-023-01155-3

Abdullah S, Barua D, Abdullah, Sk. Md A, Rabby YW (2022) Investigating the impact of land use/land cover change on present and future land surface temperature (LST) of Chittagong, Bangladesh. Earth Syst Environ 6(1):221–235. https://doi.org/10.1007/s41748-021-00291-w

Ahammed SS, Hossain MA, Abedin MZ, Khaleque MA (2015) A study of environmental impacts on the coral resources in the vicinity of the saint Martin Island Bangladesh. Int J Sci Technol Res 4(8):37–39. https://d1wqtxts1xzle7.cloudfront.net/54094396/A-Study-Of-Environmental--libre.pdf?1502255258=&response-content-disposition=inline%3B+filename%3DA_Study_Of_Environmental_Impacts_On_The.pdf&Expires=1756470535&Signature=P4x16yyLkEvgJO~oPlU8nqvsdu3jsggdbBJTPnCpiqK30GFHDZoBtrS3reu9l1UwwHDt6pbOYhq4ADHW8GT8X~yZJ2oWlWb-redIyig3ALGpcCSzfWTkNzP-X4R4zo0lIh~Ms-OkFJMr2vb6fMfL~u6gWTRdpM46TQy9g~L6XCPTy5rt79wXU7aAmVGzWVDqYrrNLHUiVsOaqZ6Bye~eJvaO0VtQ9pD85Mh6CbtGecW4vXnd0llAmK0Xhqq2r0XXquMvi4b9UefNYd00gh8MRwAr3igIFOW~t5QaKaGOIc5Mh4bVgF2D3mdxiySkijrDxtMH~bgCB99mLbHL8mSntQ__&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA

Ahmed B, Kamruzzaman M, Zhu X, Rahman M, Choi K (2013) Simulating land cover changes and their impacts on land surface temperature in Dhaka, Bangladesh. Remote Sens 5(11):5969–5998. https://doi.org/10.3390/rs5115969

Ahmed R, Mahmud KH, Hafsa B, Tuya JH, Reza E, Coalee UM (2023) Climate change impact on land use and land cover change on Saint Martin's island, Bangladesh. J Coast Conserv. https://doi.org/10.1007/s11852-023-00954-y

Alam E, Hridoy A-EE, Tusher S, Md SH, Islam AR, Md T, Islam MK (2023) Climate change in Bangladesh: temperature and rainfall climatology of Bangladesh for 1949–2013 and its implication on rice yield. PLoS One 18(10):e0292668. https://doi.org/10.1371/journal.pone.0292668

Ara S, Alif MAUJ, Islam KMA (2021) Impact of tourism on LULC and LST in a coastal Island of Bangladesh: a geospatial approach on St. Martin's Island of Bay of Bengal. J Indian Soc Remote Sens 49(10):2329–2345. https://doi.org/10.1007/s12524-021-01389-4

Avdan U, Jovanovska G (2016) Algorithm for automated mapping of land surface temperature using LANDSAT 8 satellite data. Journal of Sensors, 2016. https://doi.org/10.1155/2016/1480307

Ayyam V, Palanivel S, Chandrakasan S (2019) Coastal Ecosystems of the Tropics—Adaptive Management. Springer Singapore. https://doi.org/10.1007/978-981-13-8926-9

Biswas J, Malaker T, Mollick T (2025) Tourism and environmental change in Saint Martin island, Bangladesh: insights from remote sensing data. Remote Sens Appl Soc Environ 37:101484. https://doi.org/10.1016/j.rsase.2025.101484

Cameron RWF, Brindley P, Mears M, McEwan K, Ferguson F, Sheffield D, Jorgensen A, Riley J, Goodrick J, Ballard L, Richardson M (2020) Where the wild things are! Do urban green spaces with greater avian biodiversity promote more positive emotions in humans? Urban Ecosyst 23(2):301–317. https://doi.org/10.1007/s11252-020-00929-z

Cetin M, Ozenen Kavlak M, Senyel Kurkcuoglu MA, Bilge Ozturk G, Cabuk SN, Cabuk A (2024) Determination of land surface temperature and urban heat island effects with remote sensing capabilities: the case of Kayseri, Türkiye. Nat Hazards 120(6):5509–5536. https://doi.org/10.1007/s11069-024-06431-5

Chander G, Markham BL, Helder DL (2009) Summary of current radiometric calibration coefficients for Landsat MSS, TM, ETM+, and EO-1 ALI sensors. Remote Sens Environ 113(5):893–903. https://doi.org/10.1016/j.rse.2009.01.007

Chen XL, Zhao HM, Li PX, Yin ZY (2006) Remote sensing image-based analysis of the relationship between urban heat island and land use/cover changes. Remote Sens Environ. https://doi.org/10.1016/j.rse.2005.11.016

Chikeya CK, Ntsalaze L (2025) Determinants of household debt: a systematic review of the literature. Economies 13(3):76. https://doi.org/10.3390/economies13030076

Chowdhury SQ, Fazlul ATM, Hasan HK (1997) Beachrock in St. Martin's island, Bangladesh: implications of sea level changes on beachrock cementation. Mar Geod 20(1):89–104. https://doi.org/10.1080/01490419709388097

Chu L, Oloo F, Chen B, Xie M, Blaschke T (2020) Assessing the influence of tourism-driven activities on environmental variables on Hainan island, China. Remote Sens. https://doi.org/10.3390/rs12172813

Das J, Kabir MH, Taimur FM, Hossain M, Kumar U (2022) Evaluating governability challenges of Saint Martin's Island (SMI) in Bangladesh. World Dev Perspect 27:100434. https://doi.org/10.1016/j.wdp.2022.100434

Deng C, Wu C (2012) BCI: a biophysical composition index for remote sensing of urban environments. Remote Sens Environ 127:247–259. https://doi.org/10.1016/j.rse.2012.09.009

Dong R, Wurm M, Taubenböck H (2022) Seasonal and diurnal variation of land surface temperature distribution and its relation to land use/land cover patterns. Int J Environ Res Public Health 19(19):12738. https://doi.org/10.3390/ijerph191912738

Etikan I (2016) Comparison of convenience sampling and purposive sampling. Am J Theor Appl Stat 5(1):1. https://doi.org/10.11648/j.ajtas.20160501.11

Foga S, Scaramuzza PL, Guo S, Zhu Z, Dilley RD, Beckmann T, Schmidt GL, Dwyer JL, Joseph Hughes M, Laue B (2017) Cloud detection algorithm comparison and validation for operational Landsat data products. Remote Sens Environ 194:379–390. https://doi.org/10.1016/j.rse.2017.03.026

Guha S, Govil H, Dey A, Gill N (2018) Analytical study of land surface temperature with NDVI and NDBI using Landsat 8 OLI and TIRS data in Florence and Naples city, Italy. Eur J Remote Sens 51(1):667–678. https://doi.org/10.1080/22797254.2018.1474494

Gutman G, Huang C, Chander G, Noojipady P, Masek JG (2013) Assessment of the NASA-USGS global land survey (GLS) datasets. Remote Sens Environ. https://doi.org/10.1016/j.rse.2013.02.026

Habib M, Jubb C (2015) Poverty Alleviation with Microfinance: Bangladesh Evidence. In A. Heshmati, E. Maasoumi, & G. Wan (Eds.), Poverty Reduction Policies and Practices in Developing Asia (pp. 11–30). Springer Singapore. https://doi.org/10.1007/978-981-287-420-7_2

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

He BJ, Zhao ZQ, Shen LD, Wang HB, Li LG (2019) An approach to examining performances of cool/hot sources in mitigating/enhancing land surface temperature under different temperature backgrounds based on Landsat 8 image. Sustain Cities Soc. https://doi.org/10.1016/j.scs.2018.10.049

Hossen MF, Sultana N (2023) Shoreline change detection using DSAS technique: case of Saint Martin island, Bangladesh. Remote Sens Appl Soc Environ 30:100943. https://doi.org/10.1016/j.rsase.2023.100943

Hu Y, Hou M, Jia G, Zhao C, Zhen X, Xu Y (2019) Comparison of surface and canopy urban heat islands within megacities of Eastern China. ISPRS J Photogramm Remote Sens. https://doi.org/10.1016/j.isprsjprs.2019.08.012

Imran HM, Hossain A, Islam AKMS, Rahman A, Bhuiyan MAE, Paul S, Alam A (2021) Impact of land cover changes on land surface temperature and human thermal comfort in Dhaka City of Bangladesh. Earth Syst Environ 5(3):667–693. https://doi.org/10.1007/s41748-021-00243-4

Islam MY, Nasher NMR, Karim KHR, Rashid KJ (2023) Quantifying forest land-use changes using remote-sensing and CA-ANN model of Madhupur Sal forests, Bangladesh. Heliyon 9(5):e15617. https://doi.org/10.1016/j.heliyon.2023.e15617

Islam MY, Mohiuddin M, Hossain T, Salauddin K, Md., Farin S (2024a) Trend of heat waves in Dhaka metropolitan City and its impact on livelihood and health of exposed people. Arab J Geosci 17(8):232. https://doi.org/10.1007/s12517-024-12027-x

Islam MT, Ahmed Z, Kadir A, Ambinakudige S, Ahad MI, Hassan R, Hafiz-Al-Rezoan, Hossain MS, Pohil MAH (2024b) Influence of green space on land surface temperature in the coastal cities in Bangladesh. Environ Dev Sustain 27(2):5133–5155. https://doi.org/10.1007/s10668-024-05112-4

Kafy A-A, Abdullah-Al-Faisal, Rahman MS, Islam M, Al Rakib A, Islam MA, Khan MHH, Sikdar MS, Sarker MHS, Mawa J, Sattar GS (2021) Prediction of seasonal urban thermal field variance index using machine learning algorithms in cumilla, Bangladesh. Sustainable Cities Soc 64:102542. https://doi.org/10.1016/j.scs.2020.102542

Kafy AA, Faisal AA, Sikdar S, Hasan M, Rahman M, Khan MH, Islam, R (2020) Impact of LULC changes on LST in Rajshahi district of Bangladesh: a remote sensing approach. J Geogr Stud 3(1):11–23. https://doi.org/10.21523/gcj5.19030102

Kofarmata YI, Danlami AH (2021) A micro-level analysis of the intensity of agricultural finance supply in Nigeria: empirical evidence. SN Bus Econ 1(1):12. https://doi.org/10.1007/s43546-020-00011-y

Li L, Feng R, Xi J, Huijbens EH, Gao Y (2023) Distinguishing the impact of tourism development on ecosystem service trade-offs in ecological functional zone. J Environ Manage 342:118183. https://doi.org/10.1016/j.jenvman.2023.118183

Loveland TR, Dwyer JL (2012) Landsat: Building a strong future. Remote Sens Environ 122:22–29. https://doi.org/10.1016/j.rse.2011.09.022

Maharjan M, Aryal A, Shakya BM, Talchabhadel R, Thapa BR, Kumar S (2021) *Evaluation* of urban heat island (UHI) using satellite images in densely populated cities of South Asia. Earth. https://doi.org/10.3390/earth2010006

Marsiglio S (2017) On the carrying capacity and the optimal number of visitors in tourism destinations. Tour Econ 23(3):632–646. https://doi.org/10.5367/te.2015.0535

Massey G (2017) Social change. In The Cambridge Handbook of Sociology (Vol. 1, Issue 1). https://doi.org/10.1017/9781316418376.048

Monir MM, Rokonuzzaman M, Sarker SC, Alam E, Islam MK, Islam AR, Md. T (2023) Spatiotemporal analysis and predicting rainfall trends in a tropical monsoon-dominated country using MAKESENS and machine learning techniques. Sci Rep 13(1). https://doi.org/10.1038/s41598-023-41132-2

Nafi S, Ahmed T (2017) Sustainable tourism in saint Martin island: an observation on young tourist perception and awareness level. https://doi.org/10.9790/0837-2210117380

Ndamsa DT, Gur DM, Baye FM (2023) Household size and food consumption spending in Cameroon. Is there evidence of economies of size? SN Bus Econ 3(8):153. https://doi.org/10.1007/s43546-023-00521-5

Patton MQ (2015) Qualitative research & evaluation methods: Integrating theory and practice (Fourth edition). SAGE

Phan TN, Kuch V, Lehnert LW (2020) Land Cover Classification using Google Earth Engine and Random Forest Classifier—The Role of Image Composition. Remote Sensing 12(15), 2411. https://doi.org/10.3390/RS12152411

Priskin J (2001) Assessment of natural resources for nature-based tourism: the case of the central Coast region of Western Australia. Tour Manag 22(6):637–648. https://doi.org/10.1016/S0261-5177(01)00039-5

Rani M, Kumar P, Pandey PC, Srivastava PK, Chaudhary BS, Tomar V, Mandal VP (2018) Multi-temporal NDVI and surface temperature analysis for urban heat island inbuilt surrounding of sub-humid region: a case study of two geographical regions. Remote Sens Appl Soc Environ 10:163–172. https://doi.org/10.1016/j.rsase.2018.03.007

Rashid N, Alam JAMM, Chowdhury MA, Islam SLU (2022) Impact of landuse change and urbanization on urban heat Island effect in Narayanganj city, Bangladesh: a remote sensing-based estimation. Environ Challenges 8:100571. https://doi.org/10.1016/j.envc.2022.100571

Rashid KJ, Akter T, Imrul Kayes ASM, Yachin Islam M (2023) Exploring the Spatio-temporal patterns and driving forces of urban growth in Dhaka megacity from 1990 to 2020. In: Chatterjee U, Bandyopadhyay N, Setiawati MD, Sarkar S (eds) Urban commons, future smart cities and sustainability. Springer International Publishing, pp 375–400. https://doi.org/10.1007/978-3-031-24767-5_17

Roy B (2020) Assessment of vegetation health in Saint Martins Island, Bangladesh using remote sensing & GIS. Int J Innovations Eng Res Technol 07(07):89–93. https://repo.ijiert.org/index.php/ijiert/article/view/285

Roy DP, Wulder MA, Loveland TR, Allen CEW, Anderson RG, Helder MC, Irons D, Johnson JR, Kennedy DM, Scambos R, Schaaf TA, Schott CB, Sheng JR, Vermote Y, Belward EF, Bindschadler AS, Cohen R, Gao WB, Zhu F, Z (2014) Landsat-8: science and product vision for terrestrial global change research. Remote Sens Environ 145:154–172. https://doi.org/10.1016/j.rse.2014.02.001

Safarrad T, Ghadami M, Dittmann A (2021) & Pazhuhan (Panahandeh Khah), M. Tourism Effect on the Spatiotemporal Pattern of Land Surface Temperature (LST): Babolsar and Fereydonkenar Cities (Cases Study in Iran). Land, 10(9), 945. https://doi.org/10.3390/land10090945

Saha A, Setu S, Das S, Hossain MI, Rahman AK, Rahman MM (2024) Decision support system for community managed rainwater harvesting: a case study in the salinity-prone coastal region of Bangladesh. Heliyon 10(9):e30455. https://doi.org/10.1016/j.heliyon.2024.e30455

Scheyvens R, Momsen JH (2008) Tourism and poverty reduction: issues for small island states. Tour Geogr 10(1):22–41. https://doi.org/10.1080/14616680701825115

Shafique M, Kim R, Rafiq M (2018) Green roof benefits, opportunities and challenges – a review. Renew Sustain Energy Rev 90:757–773. https://doi.org/10.1016/j.rser.2018.04.006

Sharpley R (2012) Island tourism or tourism on islands?? Tour Recreation Res 37(2):167–172. https://doi.org/10.1080/02508281.2012.11081701

Skiniti G, Lilli M, Skarakis N, Tournaki S, Nikolaidis N, Tsoutsos T (2024) A holistic approach for tourism carrying capacity

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

estimation in sensitive ecological areas. Environ Dev Sustain 26(12):31971–31995. https://doi.org/10.1007/s10668-024-04805-0

Sodokin K (2023) Public assistance, survival, and household trust during the containment period of the first wave of the SARS-CoV-2 pandemic in Togo. J Soc Econ Dev 25(2):428–462. https://doi.org/10.1007/s40847-023-00233-4

Sultana S, Satyanarayana ANV (2020) Assessment of urbanisation and urban heat island intensities using Landsat imageries during 2000–2018 over a sub-tropical Indian city. Sustain Cities Soc 52:101846. https://doi.org/10.1016/j.scs.2019.101846

Syafii NI, Ichinose M, Kumakura E, Jusuf SK, Hien WN, Chigusa K, Ashie Y (2021) Assessment of the water pond cooling effect on urban microclimate: a parametric study with numerical modeling. Int J Technol 12(3):461. https://doi.org/10.14716/ijtech.v12i3.4126

Tania TC (2024) Socio-economic culture and tourism: a case study on Saint Martin Island of Bangladesh. Jurnal Aplikasi Manajemen Ekonomi Dan Bisnis 6(2):84–93. https://doi.org/10.51263/jameb.v6i2.148

Ullah W, Ahmad K, Tahir AA, Ullah S, Rafiq MT, Abdullah-Al-Wadud M (2025) Impacts of tourism on LULC and LST dynamics in district Buner and Shangla, Pakistan. Sci Rep 15(1):9304. https://doi.org/10.1038/s41598-025-94230-8

Upal NH (2015) Environmental Hazards and Conservation Approach to the Biodiversity and Ecosystem of the St. Martin's Island in Bangladesh. In N. J. Raju, W. Gossel, & M. Sudhakar (Eds.), Management of Natural Resources in a Changing Environment (pp. 259–269). Springer International Publishing. https://doi.org/10.1007/978-3-319-12559-6_20

USGS (2021) Landsat Collection 2 Level-2 Science Products. Fact Sheet 2021–3055. (Fact Sheet) [Fact Sheet]. https://doi.org/10.3133/fs20213055

Vermote E, Justice C, Claverie M, Franch B (2016) Preliminary analysis of the performance of the Landsat 8/OLI land surface reflectance product. Remote Sens Environ 185:46–56. https://doi.org/10.1016/j.rse.2016.04.008

Weng Q, Lu D, Schubring J (2004) Estimation of land surface temperature-vegetation abundance relationship for urban heat island studies. Remote Sens Environ 89(4):467. https://doi.org/10.1016/j.rse.2003.11.005

Wulder MA, Roy DP, Radeloff VC, Loveland TR, Anderson MC, Johnson DM, Healey S, Zhu Z, Scambos TA, Pahlevan N, Hansen M, Gorelick N, Crawford CJ, Masek JG, Hermosilla T, White JC, Belward AS, Schaaf C, Woodcock CE, Cook BD (2022) Fifty years of Landsat science and impacts. Remote Sens Environ 280:113195. https://doi.org/10.1016/j.rse.2022.113195

Yadav N (2024) Tourism, Strategies for Development (pp. 129–136)

Yanbo Q, Shilei W, Yaya T, Guanghui J, Tao Z, Liang M (2023) Territorial spatial planning for regional high-quality development – an analytical framework for the identification, mediation and transmission of potential land utilization conflicts in the Yellow River Delta. Land Use Policy 125:106462. https://doi.org/10.1016/j.landusepol.2022.106462

Zahra I (2013) Tourism and Its Impact On Cox's Bazar, Bangladesh. 15. https://doi.org/10.4316/rdt.15.42

Zhang F, Lv Y, Sarker MNI (2024) Resilience and recovery: a systematic review of tourism governance strategies in disaster-affected regions. Int J Disaster Risk Reduct 103:104350. https://doi.org/10.1016/j.ijdrr.2024.104350

Zhou Y, Guo Y, Liu Y (2020) Health, income and poverty: Evidence from China's rural household survey. Int J Equity in Health 19(1), 36.https://doi.org/10.1186/s12939-020-1121-0

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Content courtesy of Springer Nature, terms of use apply. Rights reserved.

Terms and Conditions

Springer Nature journal content, brought to you courtesy of Springer Nature Customer Service Center GmbH ("Springer Nature").

Springer Nature supports a reasonable amount of sharing of research papers by authors, subscribers and authorised users ("Users"), for small-scale personal, non-commercial use provided that all copyright, trade and service marks and other proprietary notices are maintained. By accessing, sharing, receiving or otherwise using the Springer Nature journal content you agree to these terms of use ("Terms"). For these purposes, Springer Nature considers academic use (by researchers and students) to be non-commercial.

These Terms are supplementary and will apply in addition to any applicable website terms and conditions, a relevant site licence or a personal subscription. These Terms will prevail over any conflict or ambiguity with regards to the relevant terms, a site licence or a personal subscription (to the extent of the conflict or ambiguity only). For Creative Commons-licensed articles, the terms of the Creative Commons license used will apply.

We collect and use personal data to provide access to the Springer Nature journal content. We may also use these personal data internally within ResearchGate and Springer Nature and as agreed share it, in an anonymised way, for purposes of tracking, analysis and reporting. We will not otherwise disclose your personal data outside the ResearchGate or the Springer Nature group of companies unless we have your permission as detailed in the Privacy Policy.

While Users may use the Springer Nature journal content for small scale, personal non-commercial use, it is important to note that Users may not:

1. use such content for the purpose of providing other users with access on a regular or large scale basis or as a means to circumvent access control;
2. use such content where to do so would be considered a criminal or statutory offence in any jurisdiction, or gives rise to civil liability, or is otherwise unlawful;
3. falsely or misleadingly imply or suggest endorsement, approval , sponsorship, or association unless explicitly agreed to by Springer Nature in writing;
4. use bots or other automated methods to access the content or redirect messages
5. override any security feature or exclusionary protocol; or
6. share the content in order to create substitute for Springer Nature products or services or a systematic database of Springer Nature journal content.

In line with the restriction against commercial use, Springer Nature does not permit the creation of a product or service that creates revenue, royalties, rent or income from our content or its inclusion as part of a paid for service or for other commercial gain. Springer Nature journal content cannot be used for inter-library loans and librarians may not upload Springer Nature journal content on a large scale into their, or any other, institutional repository.

These terms of use are reviewed regularly and may be amended at any time. Springer Nature is not obligated to publish any information or content on this website and may remove it or features or functionality at our sole discretion, at any time with or without notice. Springer Nature may revoke this licence to you at any time and remove access to any copies of the Springer Nature journal content which have been saved.

To the fullest extent permitted by law, Springer Nature makes no warranties, representations or guarantees to Users, either express or implied with respect to the Springer nature journal content and all parties disclaim and waive any implied warranties or warranties imposed by law, including merchantability or fitness for any particular purpose.

Please note that these rights do not automatically extend to content, data or other material published by Springer Nature that may be licensed from third parties.

If you would like to use or distribute our Springer Nature journal content to a wider audience or on a regular basis or in any other manner not expressly permitted by these Terms, please contact Springer Nature at

onlineservice@springernature.com